



DiffusionLight: a multi-agent reinforcement learning approach for traffic signal control based on shortcut-diffusion model

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Abstract

Numerous researches have shown that the Reinforcement Learning(RL) is an effective solution to solve large-scale traffic signal control(TSC) problems. However, facing multi-scenario and emergencies, cooperative control of traffic signals at multi-intersection becomes a challenging multi-agent reinforcement learning(MARL) process. In order to solve real-world problems, this paper proposes a MARL algorithm called DiffusionLight, which combines Shortcut-Diffusion Model(SDM) and Soft Actor-Critic(SAC), and a fast Diffusion model to solve the traffic signal cooperative control problem of multi-scenario and multi-intersection. DiffusionLight has the stable characteristics and powerful expression ability of the SDM as the strategy network to solve the action space, while SAC is used as the value network to better explore the solution space. Experimental results show that DiffusionLight exhibits better stability compared to the baseline algorithm in the face of multi-scenario TSC and burst data anomalies, and as well as excellent performance on multiple public datasets of grid and arterial traffic networks.

Keywords Diffusion model · Soft actor-critic · Shortcut theory · Multi-agent reinforcement learning · Burst data anomalies

1 Introduction

As the core component of an intelligent transport system, the intelligent traffic signal control(TSC) system is responsible for coordinating the traffic flow around the intersection in real time, which can effectively relieve the traffic pressure under the limited space in the city. Traditional TSC methods usually use predefined rules or fixed timing schemes [7, 35], but these manually designed control methods are unable to dynamically adjust signals according to real-time traffic flow. To overcome the challenge of changing patterns of temporal and spatial features in TSC, extensive research has shown that

RL techniques have potential applications in large-scale grid urban traffic networks [13, 14, 21, 32, 33] and arterial traffic networks [18, 41]. RL-based methods are not only able to adapt to complex urban traffic environments, but also able to achieve coordinated control among different intersections, so as to improve the overall traffic efficiency and alleviate traffic congestion more effectively.

Most existing studies assume that the traffic data of all traffic intersections are complete and continuously valid in a single scenario [20]. However, with the change of the experimental environment, the stability and adaptability of the model are greatly limited, which seriously restricts the application of intelligent TSC in reality [3]. Therefore, the focus of this paper is to explore effective and stable control scheme for large-scale urban traffic networks under multi-scenario coordinated control and data anomalies. large-scale urban grid traffic networks and arterial traffic networks have very different state characteristics. Urban traffic networks [14, 33] are affected by the complex road structure, number of intersections and traffic flow dynamics within the city. In contrast the state characteristics of arterial traffic networks [18] are more affected by the flow and topology of long-distance, arterial roads such as motorways. Therefore, a Multi-agent Reinforcement Learning (MARL) model that extracts key

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information and explores high-dimensional solution space is needed to alleviate the effects of data variations in different traffic scenarios.

In the multi-intersection TSC problem with state action spaces [18, 40], effective coordination of traffic flow at intersections has become a key problem to be solved. Facing the problem of long state space data, the current mainstream scheme is to adopt the attention mechanism [11, 18, 21], which can effectively capture key traffic information from the state space. Meanwhile, the studies [38] show that the attention mechanism can effectively capture the importance of neighboring intersections to the target intersections. In solving the multi-agent coordination problem, centralized training and decentralized execution (CTDE) is welcomed by many researches [24, 25], which adopts a unified decision-making scheme to control different executing agents and can fully utilize the local information. Research [15] has applied the above mentioned techniques to TSC with good results. However, they all have difficulty working when facing data changing in real-world scenarios, which leads to many models performing well only on specific tasks.

Compared to other models, Diffusion model (DM) [9] is well known for its high quality generation capability and powerful diversity. Recently many studies [3, 31, 42] have applied DM to RL environments, and the experimental results have proven that the introduction of DM gives more stable and excellent performance to RL algorithms, which cannot be provided by other models. The recent study [3] also apply DM to TSC, and the experiments demonstrate that DM continues to perform well against data anomalies problems. Unfortunately, all these works ignore the time consumption problem brought by DM, which makes it difficult to solve the timeliness problem of the actual TSC. Encouraged by the above works, the stability properties of one lightweight DM will bring a new solution to real-world applications of intelligent TSC.

In order to solve the above realistic problems in real-world traffic systems, we adopt the MARL algorithm framework with CTDE to ensure coordinated control while applying the stability advantage of DM. The key work of this paper is to solve the time consumption problem of the diffusion model, with acceptable traffic signal control time to enhance the stability of the algorithm in different environments and contingencies, so as to solve the problem of signal control in large-scale urban and arterial traffic. The main contributions of this paper are as follows:

1. In this paper, we firstly consider the shortcut-diffusion model (SDM) to solve the time consumption problem of the diffusion model while maintaining its advantages, so as to satisfy the practical problems of traffic signal control.
2. This paper considers the problem of coordinated control of multi-intersection TSC for grid and arterial traffic scenarios, which have distinctly different state characteristics. We propose DiffusionLight, a MARL model based on the SDM, which utilizes both the local state and the global state as the information source and effectively improves the model coordinated control capability.
3. We design the state estimation strategy in this paper, which relies on the multiple diffusion structure with SDM to evaluate the state value of the next stage and make up the anomaly data based on the estimated value. By the simple design greatly alleviate the control effect of the model under data anomalies.
4. We compare existing coordinated TSC schemes for solving multi-scenario, multi-intersection traffic signals problem on multiple publicly available datasets from grid and arterial traffic networks. At the same time, we simulate real-world intersection sensor data anomaly experiments to test the model's ability to maintain stability in realistic problems. The experimental results show that the DiffusionLight model proposed in this paper outperforms other traditional schemes in terms of convergence and stability, and exhibits excellent performance on multiple datasets collected in real traffic.

The rest of the paper is organized as follows. Section 2 discusses the research on related technologies in different TSC scenarios; Section 3 introduces the knowledge of RL applications in TSC, including transportation theory, MARL, DM, and Shortcut theory; Section 4 illustrates the design details of the DiffusionLight model and the derivation of the theory; Section 5 conducts the performance experiments of DiffusionLight in two scenarios of grid and arterial traffic, simulated data anomaly experiments of sensor failure, and ablation experiments are conducted; finally, we summarize and outlook the research work in this paper.

2 Related works

2.1 Independent TSC methods

Optimizing TSC strategies is one of the most important measures to improve transportation efficiency. Many studies have focused on applying RL to TSC problems in order to enable the system to optimize signal timing strategies by learning the traffic state in real time, thus improving the overall traffic efficiency. Independent TSC schemes have achieved significant results in numerous studies [13, 32], where each agent is responsible for signal transitions at a single intersection. In some of the studies, the agent adopts metrics from traffic theory, such as traffic pressure and waiting queue length [18, 32, 33, 41], to take the traffic flow as the current state,

while the traffic signal changes are taken as actions. The experimental results show that the deep reinforcement learning (DRL) algorithms significantly outperform the traditional TSC schemes in the independent control of intersections.

However, when the independent TSC is extended to multi-intersection, the DRL algorithms face the challenge of high-dimensional state-action space, which leads to difficulties in good control results in TSC at multi-intersection [18]. To address this challenge, several research works have designed MARL algorithms by employing centralized learning and attention mechanism in order to explore the solution space in the huge state-action space.

2.2 CTDE and attention mechanisms in TSC

Literature [6, 17] applied the centralized learning mechanism in MARL, where the centralized Critic learns the cooperation between decentralized Actors through global information. Thus, the actions performed by Actors are controlled coordinately within the constraints of the huge state-action space. Whereas, the attention mechanism captures the relationship between neighbors traffic intersections and efficiently learns the time-varying traffic flow information. [21] uses AttendLight, a generic model based on the attention mechanism in TSC, and the attention mechanism is the key technology for realizing the generic capability of AttendLight. [38] proposes AttentionLight, a model that uses an attention mechanism to capture the phase correlation of signals, to effectively solve the large-scale TSC problem. Arterial coordination is another important issue in TSC, whose main goal is to maximize arterial capacity and relieve congestion at bottleneck intersections. For this purpose, [18] proposes a MARL algorithm called MASAC in combination with Transformer [30], which captures information more efficiently and extends the solution space by efficiently exploring the arterial state-space. Literature [34] proposes a coordinated TSC model Colight based on graph neural networks, which not only considers the spatio-temporal influence of neighboring intersections on the target intersection, but also can model neighboring intersections without indicators.

2.3 Data missing TSC problem and diffusion appliance

Real-world traffic problems face data anomalies and deployment difficulties, and how to deal with anomalous data and alleviate the computational consumption of intelligent algorithms is a primary concern in the field of TSC. [36] represents the traffic movement as a simple effective pressure (EP) of vehicles queuing in traffic and solves the traffic problem through less complexity. [12] designs a spatial self-

attention module to capture dynamic spatial dependencies, which has a flow delay-aware property to explicitly model time delays in space. [20] discusses that the classic RL approaches to adaptive control do not work in the real world in the case of missing data, and it proposes a measure to compensate for this through state estimation model. However, the above schemes are still extremely unstable for TSC training of MARL at intersections in the face of the expansion of road network scale and the time-varying characteristics of traffic flow.

Diffusion model(DM) is a mathematical model capable of handling complex dynamic data and inferring system behaviors, and its introduction into the RL framework can effectively handle complex data spaces and provide stable solutions. In the field of modeling and generating complex data distributions, DM has demonstrated excellent capabilities [9, 31, 42]. Among this, [31] applies it to policy networks, which not only possess strong expressive capability, but also shows advantages such as good behavioral cloning and policy coupling. [42] proposes a new generative MARL framework MADIFF in combination with the DM, which demonstrates superior expressiveness and diversity in offline RL datasets. [3] combines DM with traffic control, designing a spatio-temporal transformer (STFormer) architecture, as well as proposing a diffusion communication mechanism (DCM) to handle the issue of missing data. The algorithm based on the DM performs better compared to other models. However, since the application of DM consumes a heavy time consumption [4, 26, 27]. The above DM-based RL schemes are difficult to be applied in large-scale urban traffic networks.

Inspired by the above studies, this paper proposes DiffusionLight, a MARL model for CTDE based on the SDM. The value network of the model adopts the entropy updating approach of soft actor-critic (SAC) [8] in order to enhance the model's ability to explore the solution space. Unlike other CTDE frameworks, this paper designs a strategy network with two execution phases.

The first phase adopts the shortcut theory [4], which mainly solves the time consumption problem of the DM, and provides a more convenient diffusion path for the DM by mapping the global observation information to the new initial hidden space through the Encoder [30] network; the second phase utilizes the conditional diffusion model, which starts diffusion in a new starting point(hidden space), and the final result is output as the action value of the strategy network. For dealing with anomalous data, we design a multiple diffusion state assessment model that relies on the SDM, which uses the Neighbor pressure to explicitly detect anomalous data and corrects the anomalous data by estimating the value.

Table 1 Variable meaning and Abbreviations of specialized terms

Symbol	Representation
$r_{m,t}$	The reward of m -th agent at time t
a^i	Output of action with i diffusion steps
Φ^i	Sum of neighboring intersection flow values divided by intersection i
$\mathcal{D}$	Replay buffer dataset
$\beta^1 \dots \beta^i \dots \beta^{N'}$	Custom hyperparameter variance table
X	Linear non-expansive mapping
ε^0	Initial estimation error
$\bar{\varepsilon}^{N',0}$	The estimation error between the reverse diffusion downward to $i = 0$
π_{θ_m}	The actor strategy of the m -th agent
$s_{m,t}$	Observed state of the m -th agent
ϵ_m	Gaussian noise in the m -th agent
ϵ_{θ_m}	Predict noise model in the m -th agent
ϕ_j	The centralized critic policy of the model, $j=1,2$
κ	Temperature coefficient
SDM	Shortcut-Diffusion Model
SAC	Soft Actor-Critic
RL	Reinforcement Learning
MARL	Multi-Agent Reinforcement Learning
DM	Diffusion Model
CTDE	Centralized Training and Decentralized Execution

3 Theoretical basis

In this subsection we present four important points. First we introduce the definitions related to transportation systems. Second we give the definitions related to MARL. Then we

briefly describe the basic theory of DM. And finally we lead to the shortcut theory which is the key definition of this paper.

3.1 Transportation related definitions

In this paper, the meanings of some variables in Eq.(3)-(31) and proper noun abbreviation are shown in Table 1. We use the three-lane single intersection shown in Fig. 1 as an example to illustrate in detail the relevant concepts in the field of transportation. These concepts can be extended to road networks with multiple intersections and arterial intersections. We define the all lanes where vehicles enter the intersection as incoming lane L_{in} , and the all lanes where vehicles leave the intersection as outgoing lane L_{out} .

- **Traffic movement:** A traffic movement is defined as the movement of vehicles from the incoming lane to the outgoing lane of the intersection, involving three movement modes: left turn, straight ahead and right turn. As Fig. 1(a) demonstrates the situation of an intersection with 8 controlled traffic signals, and Fig. 1(b) illustrates the 8 types of control signals. In most countries' traffic regulations, right-turning vehicles only need to give way when the light is red and are not restricted by traffic lights. Therefore, right turns are shown in gray in Fig. 1(a).
- **Signal phase:** A traffic Signal phase p is defined as a set of executable traffic movements. The 8-bit vector in Fig. 1(c) represents the combination of the 8 traffic signals in Fig. 1(a). Due to interference in traffic movements, traffic regulations define that some signals cannot be set to green simultaneously (e.g., signals 1 and 7 in Fig. 1(b)). We express the conflict relationship of all signals as the matrix in Fig. 1(d), where gray represents the combination of signals that are not allowed. All non-

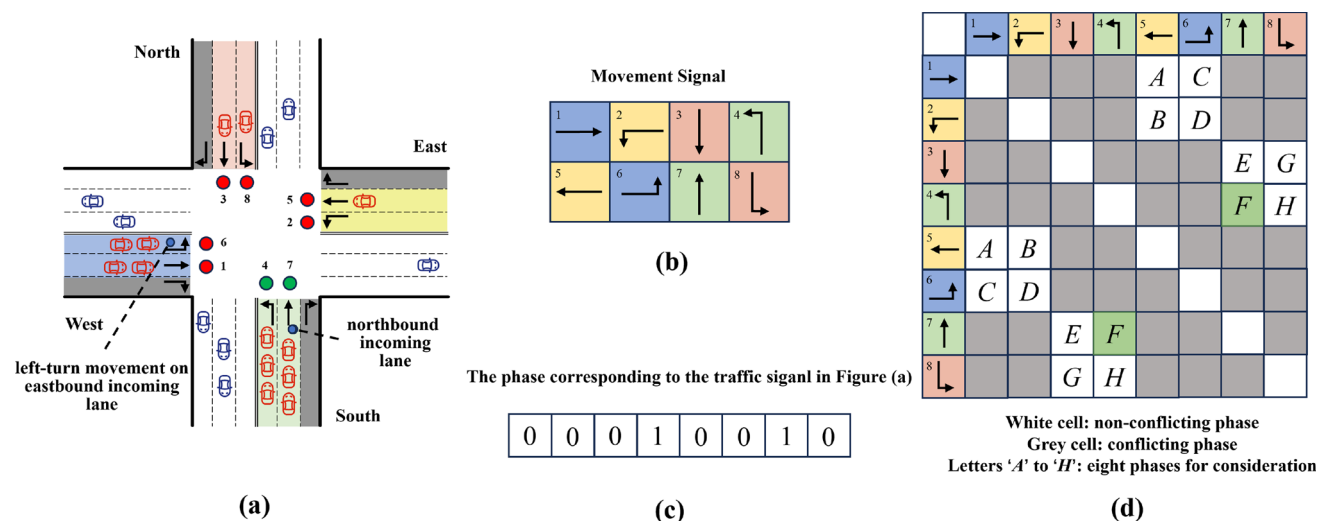


Fig. 1 Definitions of transportation theory

conflicting signals constitute 8 valid paired signal phases (e.g., labeled "A" to "H" in Fig.1(d)) and 8 single signal phases (e.g., diagonal cells in Fig. 1(d)). The traffic movements in Fig.1(a) are represented as "F" in the conflict matrix in Fig. 1(d), allowing vehicles to turn left to the west and proceed straight to the north. As in traffic movements, the use of paired signal phases brings greater benefits, the single signal phases will not be considered in this paper.

- **Intersection pressure:** the pressure value of a single intersection is defined as the difference between the total number of vehicles queuing in the incoming lane and the total number of vehicles queuing in the outgoing lane. We define the number of vehicles in the incoming lane and the number of vehicles in the outgoing lane as $s(l)$ and $s(q)$ respectively, where $l \in L_{in}$ and $q \in L_{out}$. Therefore, a single intersection pressure can be expressed by (1).

$$P_r = \left| \sum_l s(l) - \sum_q s(q) \right|, l \in L_{in}, q \in L_{out} \quad (1)$$

At the middle intersection in Fig. 2 (the intersection selected by the dotted line), the incoming lane is blue and the outgoing lane is green. The pressure value of the middle intersection is calculated according to Eq.(1) as $P_r = |(6 + 1 + 3 + 4) - (1 + 3 + 2 + 4)| = 4$. The pressure value is used to evaluate the imbalance of the quantity of vehicles, with a higher pressure value indicating a more uneven distribution of vehicles.

3.2 Multi-agent reinforcement learning method

In our problem, we consider TSC as a Markov decision process, and each agent can observe the traffic conditions at their respective intersections (Fig. 1(a)). The goal of agents is to learn a strategy that minimize the global traffic pressure value by setting appropriate phases based on partial observational information within the environment. Specifically, the key components of the TSC problem are $\langle S, O, P, R, \gamma \rangle$. The specific descriptions are as follows.

- **State space S and observation space O :** In a road network with M intersections, we set up M decision-making agents. Each agent can access partial values of the environment state $s(s \in S)$ as their current observation $o(o \in O)$. For the m -th agent's observation at time t is represented as $o_{t,m}$, which consists of the number of vehicles on each lane connected to the intersection.
- **Action set A :** In the traffic movement of TSC, at time step t , the m -th agent formulates the optimal control phase of the m -th intersection at the next Δt based on local observation $o_{t,m}$ ('A' to 'H' in Fig. 1(d)). The intersection will remain phase P_m during time step t to $t + \Delta t$.
- **State transition probability P :** Given the environment's state s_t at time t and the joint action a_t made by the agents, the environment transitions to the next state s_{t+1} based on the state transition probability $P(s_{t+1}|s_t, a_t) : S \times A_1 \times \dots \times A_t \rightarrow \Omega(S)$, where Ω represents the space of state distributions.
- **Reward function R :** We define the reward $r_{t,m}$ for agent m at moment t as the opposite of the pressure value at the m -th intersection [33]. During the interaction between the agents and the environment, by minimizing the pressure value P_m , the vehicles within the environment will be distributed evenly, effectively utilizing the green passage time. The reward $r_{t,m}$ is given by (2).

$$r_{t,m} = -P_{t,m} \quad (2)$$

- **Discount factor γ :** The discount factor γ is a parameter used to measure the relative importance of current rewards and future rewards, with a value ranging from 0 to 1. The closer γ is to 1, the more the agent emphasizes future rewards, whereas the closer γ is to 0, the more the agent prioritizes current rewards.

3.3 Diffusion process

As a powerful generative model, DM [9] describes the data generation process as a reversed forward noise process (Denoising Process). For action data $a^0 \sim p_{data}(a)$

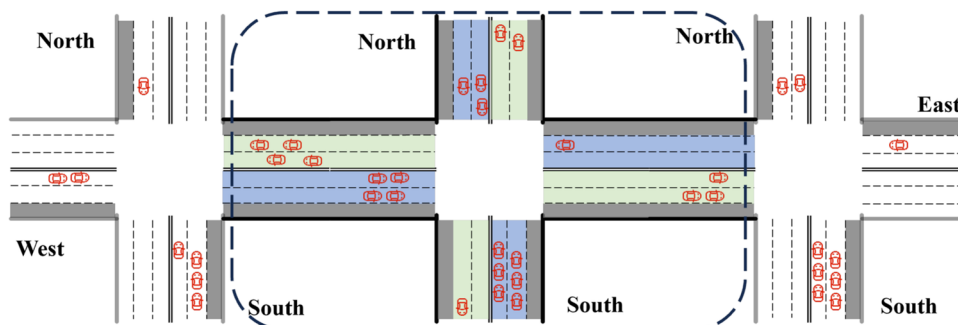


Fig. 2 Multi-intersection traffic diagram

in dataset $\mathcal{D}$, the noise process is a discrete Markov chain $a^{0:N}$, and the forward diffusion process gradually adds noise to a^0 with a predefined hyperparameter variance table $\beta^1 \dots \beta^i \dots \beta^N$, $\beta^i \in (0, 1)$, $i \in \{1, \dots, N\}$ within N steps.

As shown in Fig. 3 blue path 1, to the right indicates the reverse denoising process, and to the left represents the forward diffusion process. The forward diffusion process is a noise-adding process, and the data a^i is only related to the previous moment a^{i-1} . This process satisfies the Markov chain and is calculated using (3) and (4).

$$q(a^{1:N}|a^0) = \prod_{i=1}^N q(a^i|a^{i-1}) \quad (3)$$

$$q(a^i|a^{i-1}) = \mathcal{N}(a^i; \sqrt{1 - \beta^i} a^{i-1}, \beta^i I) \quad (4)$$

For a given initial estimate a^0 , the forward process intermediate data can be simply expressed as (5).

$$a^i = \sqrt{\bar{\alpha}^i} a^0 + \sqrt{1 - \bar{\alpha}^i} \epsilon \quad (5)$$

where $\epsilon \sim \mathcal{N}(0, I)$, $\bar{\alpha}^i = \prod_{j=1}^{i-1} \alpha^j$ and $\alpha^i = 1 - \beta^i$.

The reverse denoising process is the inverse of the forward process and can also be defined as the result of a Markov chain:

$$p_\theta(a^{0:N}) = p(a^N) \prod_{i=1}^N p_\theta(a^{i-1}|a^i) \quad (6)$$

$$p_\theta(a^{i-1}|a^i) = \mathcal{N}(a^{i-1}; \mu_\theta(a^i, i), \sum_\theta(a^i, i)) \quad (7)$$

where $p(a^N) = \mathcal{N}(a^N; 0, I)$ in Denoising Diffusion Probabilistic Models (DDPM) proposed by [9], and the mean and variance of $p_\theta(a^{i-1}|a^i)$ are given by the network $\mu_\theta(a^i, i)$ and $\sum_\theta(a^i, i)$. After training, sampled data $a \sim p(a^N)$ and the reverse diffusion chain from $i = N$ to $i = 0$

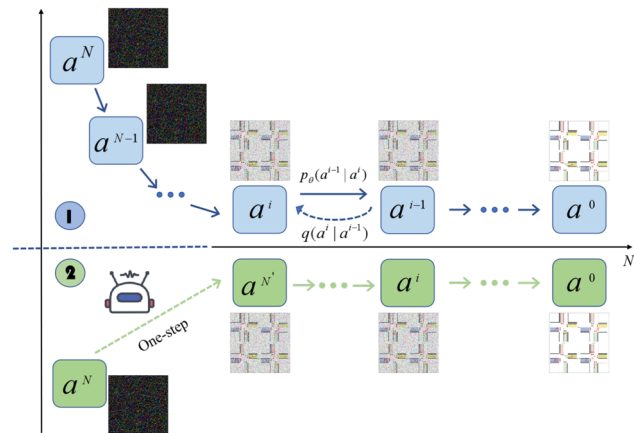


Fig. 3 Diffusion model forward diffusion and reverse denoising processes

comprise the sampling process of DM. To further enhance the capability of the policy network, DM is extended to the conditional DM [39] through condition $p_\theta(a^{i-1}|a^i, c)$.

Here we give a simple inverse process of the conditional DM and impose a data consistency operation:

$$\hat{a}^{i-1} = f(a^i, i) + g(a^i, i) \epsilon^i \quad (8)$$

$$a^{i-1} = X \hat{a}^{i-1} + b \quad (9)$$

$f(a^i, i) = \frac{1}{\sqrt{\alpha^i}} \left(a^i + \frac{\beta^i}{\sqrt{\alpha^i(1-\bar{\alpha}^i)}} \epsilon_\theta(a^i, i) \right)$, $g(a^i, i) := \sqrt{1 - \alpha^i}$, and $\epsilon \sim \mathcal{N}(0, I)$ in DDPM; b is constant and X is linear non-expansive mapping [1] as Eq.(10).

$$\|Xx - Xx'\| \leq \|a - \hat{a}\|, \forall a, \hat{a} \quad (10)$$

3.4 Shortcut path

When introducing DM into the RL algorithm to solve the TSC problem, special attention needs to be paid to the overhead of computational resources consumed by the reverse diffusion process, as shown by the blue path 1 in Fig. 3. The process will carry out N -step iterative diffusion, and the output frequency of the strategy action will be greatly reduced, which will adversely affect the real-time performance of the control system. Therefore, this paper considers adopting the shortcut theory [4] to make DM start from a new starting point and shorten the diffusion process by reducing the intermediate process, so as to obtain similar or better results. The initial starting point and diffusion end point are represented as in Definition 1.

For a given initial state a^0 , the general form of the forward diffusion N -step process can be defined as (11).

$$a^N = k^N a^0 + b^N \epsilon \quad (11)$$

where $\epsilon \sim \mathcal{N}(0, I)$, $k^N = \sqrt{\bar{\alpha}^N}$ and $b^N = \sqrt{1 - \bar{\alpha}^N}$ are the simple parameterized representations of (5)

Definition 1 Let $a^0 \in \mathbb{R}^n$ and $\tilde{a}^0 \in \mathbb{R}^n$ represent the true initial state and estimated initial state of the action data distribution respectively, then the initial estimation error is $\varepsilon^0 = \|a^0 - \tilde{a}^0\|^2$. Correspondingly, $a^{N'}$ and $\tilde{a}^{N'}$ are obtained based on (11), then the estimation error after the forward diffusion is as follows:

$$\begin{aligned} \bar{\varepsilon}^{N'} &:= \mathbb{E} \|a^{N'} - \tilde{a}^{N'}\|^2 \\ &= (k^{N'})^2 \varepsilon^0 + 2(b^{N'})^2 n \end{aligned} \quad (12)$$

Theorem 1 According to the stochastic contraction property of the stochastic difference equation [22, 23], considering the reverse diffusion process using (8) and (9), then we have:

$$\bar{\varepsilon}^{N',0} \leq \frac{2C\tau}{1-\lambda^2} + \lambda^{2N'} \bar{\varepsilon}^{N'} \quad (13)$$

Where $\bar{\varepsilon}^{N',0}$ represents the estimation error between the reverse conditional diffusion path spreading downward to $i = 0$, and $\tau = \frac{\text{Tr}(A^T A)}{n}$. In addition, the closed form expression for the contraction rate λ and the constant C in DDPM is as follows:

$$\lambda = \max_{i \in [N']} \sqrt{\alpha^i} \left(\frac{1 - \bar{\alpha}^{i-1}}{1 - \bar{\alpha}^i} \right) \quad (14)$$

$$C = n \max_{i \in [N']} (1 - \alpha^i) \quad (15)$$

The main results obtained above demonstrate the existence of the shortcut path for the acceleration. Please refer to the Appendix for specific proof.

Theorem 2 *Shortcut path.* For any $0 < \mu < 1$, there exists a minimum value N' ($N' < N$) such that $\bar{\varepsilon}^{N',0} \leq \mu \varepsilon^0$. The value of N' is satisfied $2\log(\frac{4n}{\mu \varepsilon^0}) \leq N' \beta^{N'} \leq \frac{\mu \varepsilon^0}{4n\tau}$, that N' decreases as ε_0 gets smaller.

Theorem 1 states that the conditional reverse diffusion process shrinks exponentially. On this basis, **Theorem 2** demonstrates that the reverse process of DM can obtain the same estimated result in a shorter number of denoising steps. Therefore, As shown in Fig. 3 green path 2, we introduce a *One-Step* network to provide a more convenient diffusion starting point for the DM by learning a new initial state $a^{N'} (N' < N)$.

4 Method

This section first introduces the overall structure of DiffusionLight, next describes how Encoder can be used as the *One-Step* network to realize Shortcut-Diffusion, then explains in detail how the DM outputs actions and estimates the next state based on the observations, and finally constructs the complete CTDE framework using Critic networks with entropy updating. In this section, due to the different types of time steps for DM and RL, we will use the superscript $i \in 1, \dots, N'$ ($N' < N$) to describe the diffusion time step (N' indicates adopting the shortcut theory, and N means do not employ it), and the subscripts $t \in 1, \dots, T$ and $m \in 1, \dots, M$ indicate the reinforcement learning trajectory time step and the number of the intelligentsia respectively. The pseudocode for DiffusionLight is shown in **Algorithm 1**.

4.1 The overall construction of the model

DiffusionLight adopts the CTDE [7] framework, and we employ the entropy updating approach of SAC as the basis of the value network, while designing the strategy network with two phases to enable DiffusionLight to better utilize both global and local observations in the multi-agent cooperative environment to improve the coordination of the system. The overall framework of DiffusionLight is shown in Fig. 4.

The strategy network of the algorithm consists of two phases is shown as the red dotted line in the Fig. 4, including *One-Step* shortcut and DM decision process.

The first stage: The application of Shortcut theory to the DM is to reduce the intermediate process through *One-Step* operation by mapping the global observation into a hidden space and requiring ε^0 to be as small as possible. Therefore, the *One-Step* operation requires extracting useful information from the complex and high-dimensional state space and mapping it nonlinearly to a new diffusion starting point $a^{N'}$. In order to completely take advantage of the attention mechanism in data extraction, this paper adopts Encoder as the first stage for data extraction.

The second stage: Through Shortcut theory, the DM is sampled from the hidden space of the first stage and diffuses N' times through the Denoise network under the condition of local observation. The inputs of Denoise include the current number of diffusion steps, local observations, and the output of the last Denoise iteration. DeNoise adds the memory loop network GRU [5] to the original, which prevents anomalous data by estimating the next stage of observation from previous experience. The Diffusion process of DM and the Denoise network structure are abbreviated as DF and DN in Fig. 4.

4.2 Encoder as one-step network

In the strategy network, the hidden space is learned by Encoder network to obtain more efficient diffusion points. Specifically, to enhance the cooperation between agents, the global state S_t of time t is encoded into the hidden space $a^{N'} (N' < N)$ by D -layer Encoder. In the Encoder module, S_t is encoded by a three-layer MLP network, fusing the Sinusoidal position encoding X_{pos} vector into the hidden space:

$$X_{emb} = MLP(S_t) + X_{pos} \quad (16)$$

Consider the self-attention of a single head as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}} V\right) \quad (17)$$

where $Q, K, V \in \mathbb{R}_n$, $\sqrt{d_k}$ converts the attention matrix into a standard normal distribution, n is the dimension of the

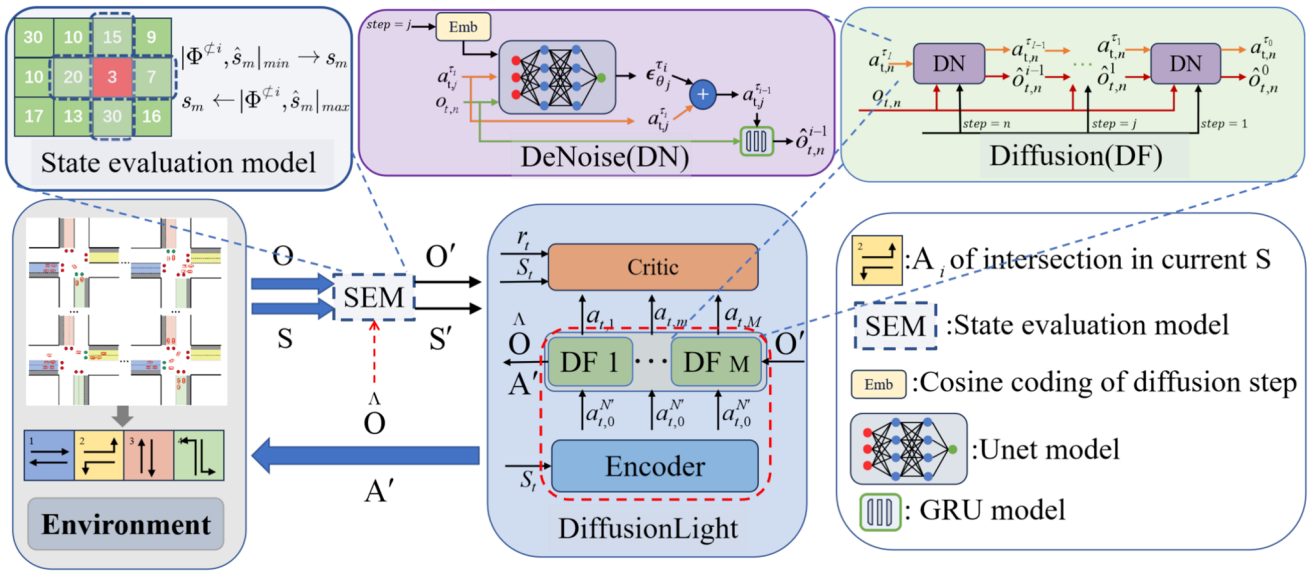


Fig. 4 The overall construction of the DiffusionLight

Algorithm 1 DiffusionLight.

Require: Actor network parameters θ_m including Encoder and Diffusion Models, Critic network parameters ϕ_j ($j = 1, 2$), temperature parameter α , agent set v , soft update parameter ξ , replay buffer $\mathcal{D}$, time horizon T and max epochs K .

Ensure: $\{\theta_m, \phi_j$ ($j = 1, 2$), $\alpha\}$. Set parameters for target Critic network ϕ_{jtar} ($j = 1, 2$): $\phi_{jtar} \leftarrow \phi_j$;

```

1: for  $k = 1 \rightarrow K$  do
2:   for  $t = 1 \rightarrow T$  do
3:     observe  $O_t, S_t$ ;
4:     The observations are modified by the SEM model  $\triangleright$  in data missing experiment
5:     Encoder outputs diffusion initial latent variable  $a^{N'}$  based on (18);
6:     DMs diffuse from  $a^{N'}$  with  $o_{m,t}$  as the condition, and calculate  $a_m^0$  based on (22);
7:     execute action set  $a_m^0$  in the environment;
8:     store  $\{S_t, o_{m,t}, a_{m,t}, S_{t+1}, o_{m,t+1}\}$  in  $\mathcal{D}$ ;
9:   end for
10:  calculate the loss of the actor according to (28);
11:  update Critic parameters  $\phi_j$  based on (29) and (30);
12:  adjust temperature parameter  $\kappa$  based on (31);
13:  soft update:
14:   $\phi_{jtar} \leftarrow \xi \phi_j + (1 - \xi) \phi_{jtar}$  ( $j = 1, 2$ );
15: end for

```

input vector X_{pos} . Q, K, V follows the definition of Transformer [30].

Finally, the observations are mapped into the hidden space $a^{N'}$ through Feed-forward networks.

$$a^{N'} = \text{Relu}((X_{emb} + \text{Attention}(Q, K, V) * W)) \quad (18)$$

As shown in Fig. 3 green path 2, the process of exploring the shortcut of DM is to encode the global observation into a

new initial diffusion state $a^{N'}$ by Encoder, which is the core point of **Theorem 2**. Not only does this allow for a more efficient diffusion process, but also enables multiple DMs to share the same initial state. As DMs have common access to global information, this enhances the ability to coordinate among Actors.

4.3 Diffusion model

The action output of strategy θ_m of the second stage in the m -th agent is defined as the reverse diffusion process of DM. The steps i to $i - 1$ of the reverse diffusion process of strategy θ_m is a denoising process, where the noise is parameterized to the probability function $p_{\theta_m}(a_m^{i-1} | a_m^i, s_m)$ under the given observation condition s_m and obeying $\mathcal{N}(a_m^{i-1}; \mu_{\theta_m}(a_m^i, s_m, i), \sum_{\theta_m}(a_m^i, s_m, i))$. And the mean square error and mean value are calculated by (19) and (20) respectively.

$$\sum_{\theta_m}(a_m^i, s_m, i) = \beta^i I \quad (19)$$

$$\mu_{\theta_m}(a_m^i, s_m, i) = \frac{1}{\sqrt{\alpha^i}}(a_m^i - \frac{\beta^i}{\sqrt{1 - \tilde{\alpha}^i}} \epsilon_{\theta_m}(a_m^i, s_m, i)) \quad (20)$$

Where, $\beta^1 \dots \beta^i \dots \beta^{N'}$, $\beta^i \in (0, 1)$, $i \in (1 \dots N')$ represents the predefined hyperparameter variance table, $\tilde{\alpha}^i = \prod_{j=1}^{i-1} \alpha^j$ and $\alpha^i = 1 - \beta^i$. ϵ_{θ_m} is the noise prediction model and all DMs in strategy π_{θ_m} ($m \in 1 \dots M$) use the same parameters. The reverse process of the conditional DM samples from the output value $a^{N'}$ of the Encoder network. The m -th action

strategy in RL is modeled as (21).

$$\begin{aligned}\pi_{\theta_m}(a_m|s_m) &= P_{\theta_m}(a_m^{0:N'}|s_m) \\ &= a^{N'} \prod_{i=1}^{N'} p_{\theta_m}(a_m^{i-1}|a_m^i, s_m)\end{aligned}\quad (21)$$

Where a_m^0 represents the output result of the m -th DM, and the action set $a^0 = \{a_1^0 \dots a_m^0 \dots a_M^0\}$, $m \in \{1 \dots M\}$ is the action output of the RL network. By parameterizing the reverse diffusion chain represented by strategy π_{θ_m} , the diffusion process is given by (22).

$$\begin{aligned}a_m^{i-1} | a_m^i &= \frac{a_m^i}{\sqrt{\alpha^i}} - \frac{\beta^i}{\sqrt{\alpha^i(1-\bar{\alpha}^i)}} \epsilon_{\theta_m}(a_m^i, s_m, i) \\ &+ \sqrt{\beta^i} \epsilon_m, \epsilon_m \sim \mathcal{N}(0, I), \text{ for } i = N', \dots, 1\end{aligned}\quad (22)$$

In order to better train the noise prediction model ϵ_{θ_m} in the conditional DM, this paper introduces a simplified objective loss function inspired by the research of [9]. The objective function of ϵ_{θ_m} in the strategy π_{θ_m} is (23).

$$\begin{aligned}\mathcal{L}_\epsilon(\theta_m) &= \mathbb{E}_{i \sim \mathcal{U}, \epsilon \sim \mathcal{N}(0, I), s_{m,t} \sim \mathcal{D}, a_{m,t} \sim \pi_{\theta_m}} [\|\epsilon_m \\ &- \epsilon_{\theta_m}(\sqrt{\bar{\alpha}^i} a_{m,t} + \sqrt{1-\bar{\alpha}^i} \epsilon_m, s_{m,t}, i)\|^2]\end{aligned}\quad (23)$$

Here, $\mathcal{U}$ is the uniform distribution of the discrete set as $i \in \{1, \dots, N'\}$ ($N' < N$) and $\mathcal{D}$ represents the data collected by action strategies π_θ in the replay experience pool through online learning.

The prediction of the next observation is made based on the current observation s_m^i and the estimated action a_m^i through the GRU, and the estimation function is (24).

$$\hat{s}_m^{i+1} = \text{Norm}(\text{GRU}(s_m^i, a_m^i)) \quad (24)$$

For the detection of anomalous data, we use the sum of the flow values of neighboring intersections $\Phi^{\mathcal{Z}^i}$ and the estimated state $\hat{s}_m$ for evaluation as in (25).

$$s_m = \begin{cases} \min(\hat{s}_m, \Phi^{\mathcal{Z}^i}), & \Phi^{\mathcal{Z}^i} > s_m \\ \max(\hat{s}_m, \Phi^{\mathcal{Z}^i}), & \Phi^{\mathcal{Z}^i} < s_m \end{cases} \quad (25)$$

The missing traffic flow can be calculated simply by the difference between neighboring intersections $\Phi^{\mathcal{Z}^i} \neq s_m$. For intersections with missing data, we use the conservative value of (25) as the new state.

We use L2 loss to align the actual and estimated observations with the objective function of (26).

$$\mathcal{L}_\wedge = \sum_{m=1}^M (s_m^i, \hat{s}_m^i)^2 \quad (26)$$

The sum of the expected reward values in standard RL is usually expressed as $\sum_t \mathbb{E}_{(s_t, a_t) \sim \rho_\pi} [r(s_t, a_t)]$, and the goal strategy update formula is designed to maximize entropy at the same time. In SAC [8] algorithm, based on the idea of Double DQN [29], two value network functions $Q(\phi_1$ and $\phi_2)$ are used simultaneously to alleviate the problem of over-estimation of Q value. The loss function of the strategy π_{θ_m} is simplified by the KL divergence as (27).

$$\begin{aligned}\mathcal{L}_Q(\theta_m) &= \mathbb{E}_{s_{m,t} \sim \mathcal{D}, a_{m,t} \sim \pi_{\theta_m}} [\kappa_t \log(\pi_{\theta_m}(a_{m,t}|s_{m,t})) \\ &- \min_{j=1,2} (Q_{\phi_j})(s_{m,t}, a_{m,t})]\end{aligned}\quad (27)$$

Where κ_t is the temperature coefficient. DiffusionLight introduces DMs into Actors, so the loss of m -th Actor in our algorithm is updated with (28).

$$\mathcal{L}(\theta_m) = \mathcal{L}_\epsilon(\theta_m) + \mathcal{L}_Q(\theta_m) + \mathcal{L}_\wedge \quad (28)$$

4.4 Soft actor-critic

[8] proposes a off-policy algorithm based on the maximum entropy RL framework, known as the Soft Actor-Critic (SAC). The goal of the value network is to maximize expected return and entropy value at the same time. The SAC algorithm encourages agents to explore the environment more deeply, rather than abandoning paths with lower current reward values. Moreover, SAC can capture multiple optimal behavioral patterns when there are several actions with equal probabilities, leading to improved performance. We optimize the Critic network by minimizing the loss function through (29).

$$\begin{aligned}J(\phi_j) &= \sum_{m=1}^M \mathbb{E}_{(s_{m,t}, a_{m,t}) \sim \mathcal{D}} [\frac{1}{2} (\pi_{\theta_m}(s_{m,t})^T Q_{\phi_j}(s_{m,t}) \\ &- y(r_{m,t}, s_{m,t+1}))^2], j = 1, 2\end{aligned}\quad (29)$$

where $\mathcal{D}$ denotes the replay buffer, $Q_\phi(\cdot)$ means the Q value computed by the Critic network, and $y(\cdot)$ represents the Q value computed by the target Critic network. The target value $y(\cdot)$ is given by (30).

$$\begin{aligned}y(r_{m,t}, s_{m,t+1}) &= r_{m,t} + \pi_{\theta_m}(s_{m,t})^T \\ &\times (Q_{\phi_{tar}}(s_{m,t+1} - \kappa_t \log \pi_{\theta_m}(s_{m,t+1})))\end{aligned}\quad (30)$$

In (29), we refer to the work of [8] to design the temperature coefficient κ_t automatic temperature method of (31).

$$\begin{aligned}\kappa_t^* &= \frac{1}{M} \arg \min_{\kappa_t} \sum_{m=1}^M \mathbb{E}_{a_{m,t} \sim \pi_{m,t}^*} \\ &[-\kappa_t \log \pi_{m,t}^*(a_{m,t} | s_{m,t}; \kappa_t) - \kappa_t \bar{\mathcal{H}}]\end{aligned}\quad (31)$$

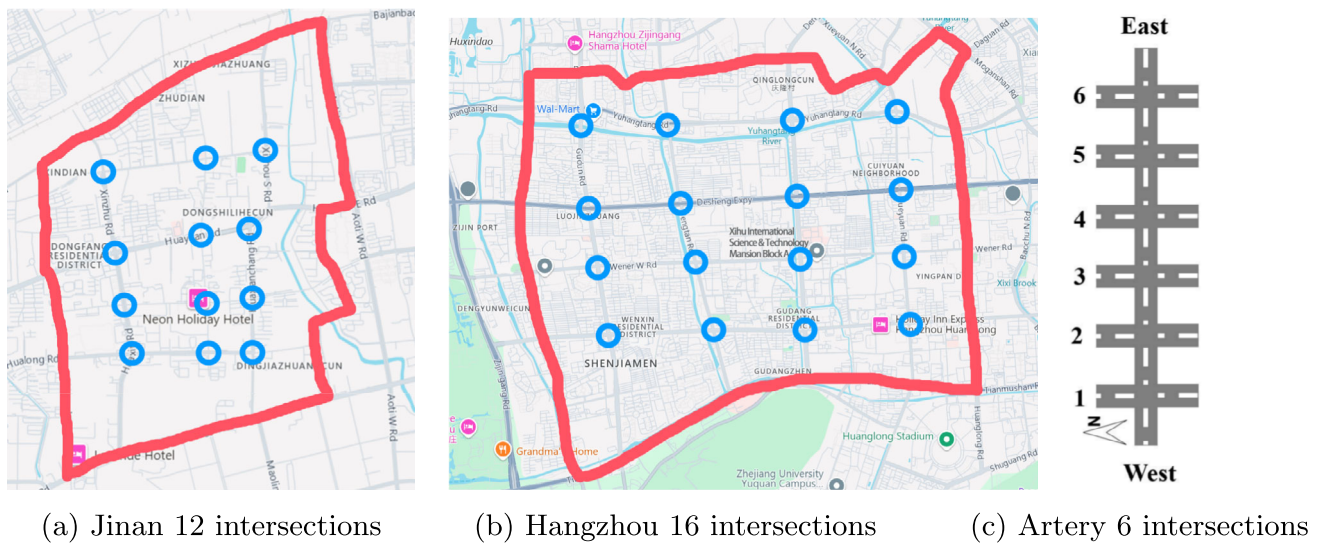


Fig. 5 (a)(b) is the grid traffic network, where the red polygon represents the selected area and the blue dot represents the selected traffic intersection. (c) 6×1 artery traffic network

Where $\bar{H}$ is a constant vector representing the target entropy of the m -th Actor.

5 Experiment

In this section, we first introduce the datasets, comparison algorithms, and evaluation metrics, and then carry out coordinated control experiments on the grid and arterial road networks, while we observe the metrics of global pressure and local pressure to provide an in-depth analysis of the model. After that, we set up traffic data anomaly experiments to simulate real traffic situations and evaluate the network's resistance to risky data under multiple anomalies. Finally, we employ ablation experiments to analyze DiffusionLight's networks and parameters individually. We conduct multiple intersections TSC experiments on the experimental platform LibSignal [19] including the simulator CityFlow [37]. We conduct our experiments On a 3070ti NVIDIA graphics card. The best results of all comparison experiments will be exhibited in bold font.

5.1 Datasets

We conduct experiments using two types of traffic networks: grid and arterial road networks, as shown in Fig. 5.

- **Real-world datasets:** We utilize two datasets collected from real traffic in Hangzhou and Jinan [16]. These datasets are obtained from traffic networks exported from Google Maps as shown in Fig. 5. Traffic data is collected by roadside surveillance cameras, which record the trajectories of vehicles as they pass by. Each record in the

datasets includes time, camera ID, and vehicle information. As shown in Fig. 5(a) displays the traffic network of Dongfeng Street in the Jinan dataset, containing 12 intersections in a 3×4 grid format. Figure 5(b) presents the traffic network of Gudang Street in the Hangzhou dataset, which consists of 16 intersections arranged in a 4×4 grid format.

- **Synthetic datasets:** This Synthetic $_{6 \times 1}$ is synthesized by statistical analysis of real traffic patterns in Jinan and Hangzhou, and is dedicated to conducting experiments on arterial road network. In our experiments, we test the DRL algorithm using the public datasets¹ of arterial network traffic. Figure 5(c) illustrates the traffic network used for the synthetic dataset, consisting of 6 intersections in a 6×1 arterial road network. To validate the algorithm's performance under different traffic conditions, we select both flat-hour and peak-hour traffic datasets.

5.2 Comparison methods

In order to fully validate the effectiveness of the proposed algorithm in this paper, it is compared in experiments with traditional and RL-based TSC schemes. To ensure the fairness of the comparison, the interval between the actions in each group of experiments is set to 10 seconds, and the time of each round of simulation lasts 60 minutes. The traditional TSC schemes are preset processes without training, the results of which are presented in the table of experimental results. The training process based on RL algorithm does not involve any pre-training model. The training parameters of

¹ https://github.com/wingsweihua/presslight/tree/master/data/template_lsr

Table 2 Comparison experimental parameters of the TSC schemes based on RL

Hyperparameter	GA	PressLight	CoLight	MpLight	AttentionLight	DiffLight	Ours
Lr(α)	$\times$	3e-4	3e-4	3e-4	3e-4	3e-4	3e-4
Gamma(γ)	$\times$	0.95	0.95	0.95	0.95	0.95	0.95
Episode	200	200	200	200	200	200	200
Time step(t)	10s	10s	10s	10s	10s	10s	10s
Horizon(T)	3600s	3600s	3600s	3600s	3600s	3600s	3600s
Memory capacity	5000	5000	5000	5000	5000	5000	5000
Batch size	64	64	64	64	64	64	64
Init entropy($\bar{\mathcal{H}}$)	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$-\log 0.1$

all comparison algorithms are set in Table 2. DiffusionLight use the 'Ours' parameter settings. This subsection involves comparison experiments with multiple models, and we map the models to the colors as in Fig. 6.

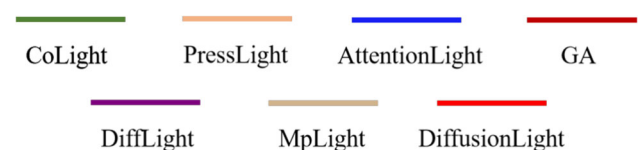
- Genetic Algorithm(GA) [10]²: It is an algorithm inspired by biological evolutionary processes for solving complex optimization and search problems.
The traditional TSC schemes used for comparison are as follows:
- FixedTime [35]: Fixed-time TSC uses a predefined fixed period length and phase timing plan, and has been widely applied in TSC.
- MaxPressure [23]: MaxPressure TSC is an advanced control method, which is widely used in network level signal control. It greedily selects the phase based on the waiting queue length of upstream and downstream traffic flow, in order to minimize the traffic pressure of a single intersection and achieve the maximum throughput of the entire traffic network.
- SOTL [35]: Self-organizing TSC is an adaptive method that controls traffic signals by manually adjusting the threshold of the number of waiting vehicles. When the number of waiting vehicles exceeds the threshold, the traffic signal will change.
The RL-based TSC schemes used for comparison are:
- PressLight [33]: A RL-based TSC algorithm that combines the concept of pressure from transportation theory in its design. It has shown superior performance in the problem of TSC.
- CoLight [34]: A TSC approach that combines graph convolutional neural networks and RL with the DQN algorithm, which models information from neighboring intersections as a graph structure to optimize queue lengths.
- MpLight [2]: A TSC scheme using parameter-sharing RL DQN algorithm. It combines RL methods and traffic

phase conflict theory, and uses phase pressure p and current phase s as observations, effectively addressing TSC problems in large-scale road networks.

- AttentionLight [38]: A RL algorithm applying attention mechanism to extract effective information from a high-dimensional state space has excellent results in TSC problems.
- DiffLight [3]: An on-policy network combining the DM for accurate action estimation and simultaneously performing the detection of anomalous data can ensure the stability of the TSC problems.

5.3 Evaluation index

- 1) Average travel time (ATT)(unit: s) : The average travel time is the average sum of the time taken by all vehicles from the initial point to the end point, which can directly reflect the impact of traffic signal switching strategies on vehicle running.
- 2) Average queue length (AQL)(unit: vehicle) : The average queue length is the average sum of all vehicles queuing in the incoming lane, which can effectively reflect the traffic congestion.
- 3) Average intersection pressure (AIP) and Average single intersection pressure(ASIP): The Average intersection pressure is the difference between the sum of all incoming and outgoing vehicles, which can reflect the ability of traffic signal switching strategies to control traffic congestion. Here we evaluate both the global Average intersection pressure and the Average single intersection pressure of the highest flow. The id and flow for the maximum flow intersections are shown in Table 3.

**Fig. 6** The model corresponds to the color curve

² <https://github.com/ahmedfgad/GeneticAlgorithmPython>

Table 3 Maximum traffic flow at the intersection of the road network (vehicles/hour)

Item	Flat _{6×1}	Peak _{6×1}	Jinan _{3×4}	Hangzhou _{4×4}
Id	2_1	5_1	1_4	1_1
Flow	1321	1306	1236	1428

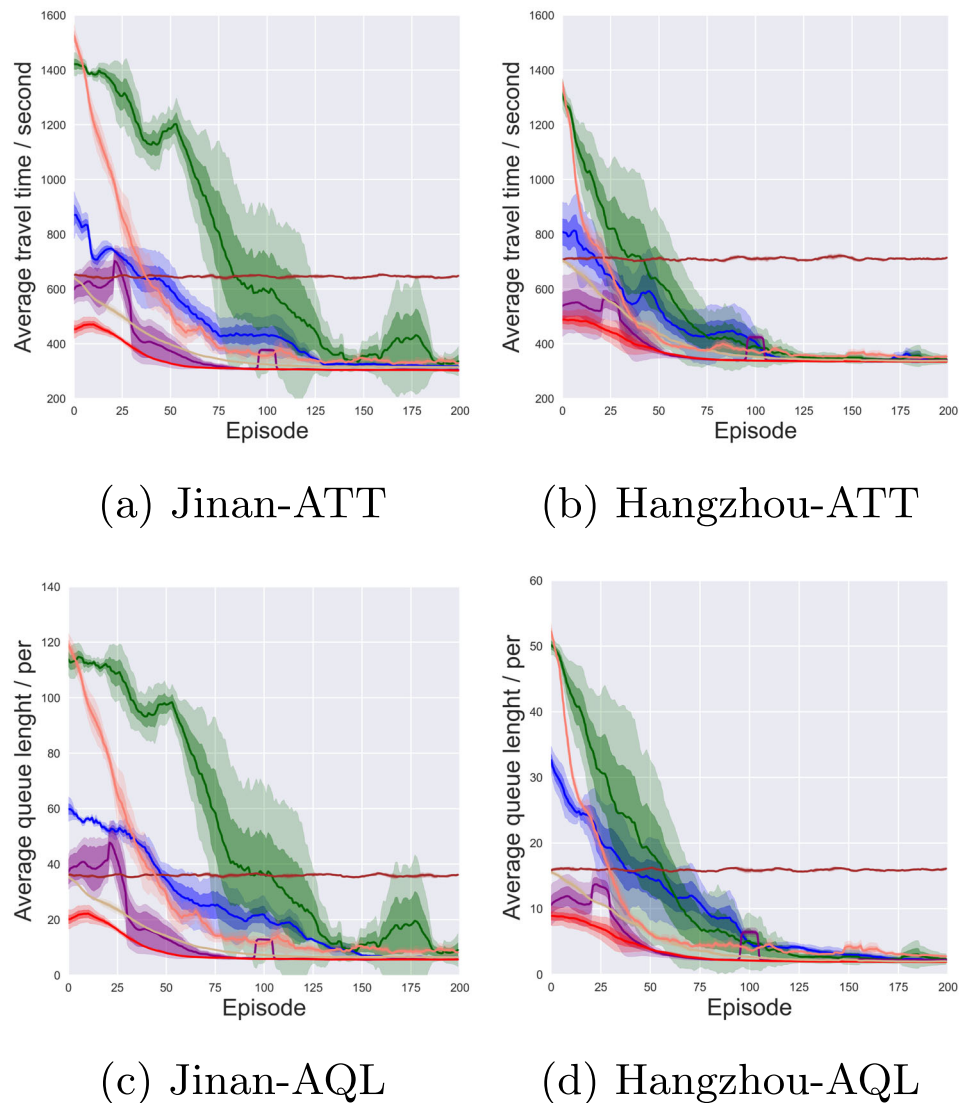
5.4 Grid transportation network performance comparison

The grid network exhibits strong topological structure characteristics. When there is congestion at a certain intersection, it will seriously affect the transportation state of neighboring traffic intersections, which makes the transportation capacity of the whole network worse. To verify the performance of different models in solving grid networks problems, we carry out experiments on Jinan_{3×4} and Hangzhou_{4×4} grid

networks. At the same time, in order to ensure the fairness of the experiment, each group of experiment is conducted 5 times to take the average value.

Figure 7 shows the ATT and AQL metrics for different algorithms for grid TSC, respectively. Due to the complexity of coordinated TSC, GA is unable to learn an effective control strategy. PressLight, CoLight, AttentionLight, and DiffLight all have significant waves, which can have serious consequence for the control task. Compared to other models, both MpLight and DiffusionLight have excellent convergence properties and stability. And in particular, DiffusionLight has an even better convergence rate.

Figure 8 demonstrates the ability of different models to regulate the AIP and the ASIP. As shown in Fig. 8(a) and (b), the regulation curves of CoLight and PressLight are very unstable, which explains the floating problem in both ATT and AQL metrics. Specifically for the intersections with the highest flows in Table 3, Fig. 8(c) and (d) show that CoL-

Fig. 7 Performance of different algorithms under the ATT and AQL metrics

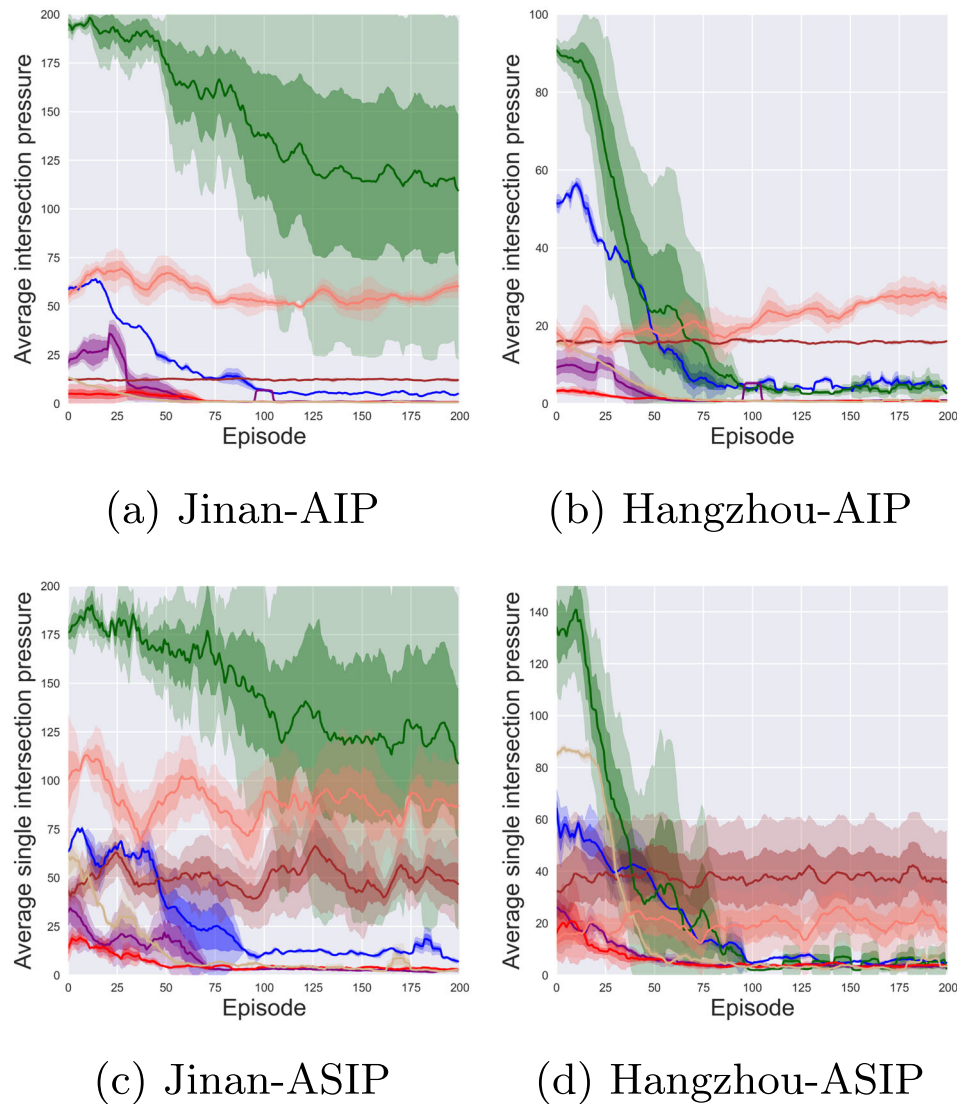


Fig. 8 Performance of different algorithms under the AIP and ASIP metrics

ight, PressLight, AttentionLight, and MpLight still fluctuate despite model convergence. In comparison, both DiffLight and DiffusionLight show better stability.

As can be observed from Table 4, the heuristic algorithm has no advantage over the traditional control methods. The RL-based PressLight algorithm does not have outstanding results compared to the traditional control algorithms Max-Pressure and SOTL, while the AttentionLight, CoLight and MpLight networks, which are well-designed, all present certain advantages in terms of optimal results. The algorithm DiffLight based on the DM framework presents an improvement over other methods. The DiffusionLight proposed in this paper achieves significantly better results than the comparison algorithms in several metrics.

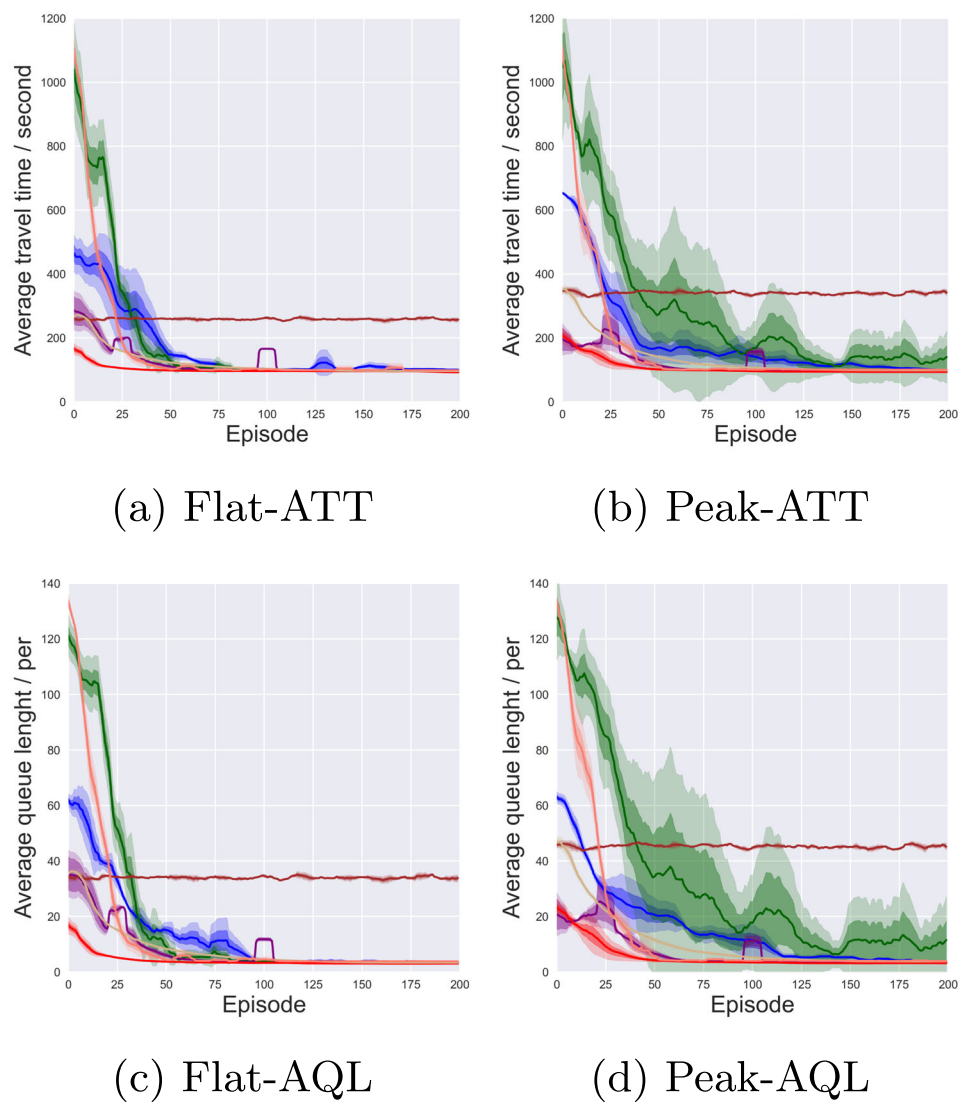
5.5 Arterial transportation network performance comparison

Unlike the grid network, arterial network needs to prioritize the control of arterial roads, while minimizing the pressure on branch lines to increase the traffic of the entire network. The synthetic dataset is used in the arterial traffic network, which includes both $\text{Flat}_{6 \times 1}$ and $\text{Peak}_{6 \times 1}$ traffic flows. We also compare the traditional TSC algorithms and the RL-based algorithms to verify the generalization of the model. The experimental results show that our algorithm improves in stability and convergence speed.

In the comparison experiments in the arterial traffic road network, CoLight shows strong fluctuations in the $\text{Peak}_{6 \times 1}$

Table 4 The results of grid transportation network comparison are averaged by 5 times for each model

Models	Hangzhou _{4×4}				Jinan _{3×4}			
	ATT	AQL	AIP	ASIP	ATT	AQL	AIP	ASIP
FixedTime	575.556	12.369	×	×	425.314	17.868	×	×
MaxPressure	365.063	3.597	×	×	328.963	8.178	×	×
SOTL	354.125	2.723	×	×	333.584	8.168	×	×
GA	701.998	15.643	15.316	31.935	638.579	35.265	11.60	39.244
PressLight	350.000	2.704	15.315	3.073	328.726	8.265	50.317	68.730
CoLight	342.142	2.285	2.450	2.240	325.524	7.894	111.046	110.494
MpLight	337.437	1.947	0.534	3.019	314.391	6.441	0.625	2.707
AttentionLight	338.094	2.143	3.449	3.035	316.896	6.436	3.946	7.227
DiffLight	335.124	1.907	0.477	2.001	306.892	6.117	0.599	2.125
DiffusionLight	334.445	1.867	0.376	1.555	302.010	5.577	0.541	1.991

Fig. 9 Performance of different algorithms in ATT and AQL metrics

dataset of Fig. 9(b) and (d) peak hour, which indicates that CoLight lacks the ability to deal with the state distribution of complex data. Contrary to the experimental results of the grid road network, PressLight with employs an independent MARL framework, outperforms AttentionLight, CoLight, MpLight, and DiffLight, in terms of both convergence speed and stability. This demonstrates that network frameworks well-designed for a certain type of task are instead less capable on other tasks, with low generalization of the model. For convergence, DiffusionLight reaches convergence before 50 rounds of iterations in both arterial tasks compared to the comparison algorithms, and the other models all exceed 75 rounds before they begin to converge. For stability, AttentionLight, CoLight, and DiffLight have significant fluctuations

in their curves, while both MpLight and DiffusionLight have excellent stability in the arterial task.

Figure 10 shows the regulation ability of AIP and SAIP for different models. In Fig. 10, the independent PressLight shows better stabilization than the coordinated algorithms such as CoLight, MpLight, and DiffLight, which illustrates that the well-designed structures perform larger differences in different scenarios. In contrast, DiffusionLight extracts information through the attention mechanism and performs optimally on both AIP and ASIP metrics for the arterial task.

As can be seen from Table 5, GA is difficult to learn an effective control strategy, and the expandability and stability of CoLight and AttentionLight on the arterial task are very fragile with significant floating. In comparison, the algorithm

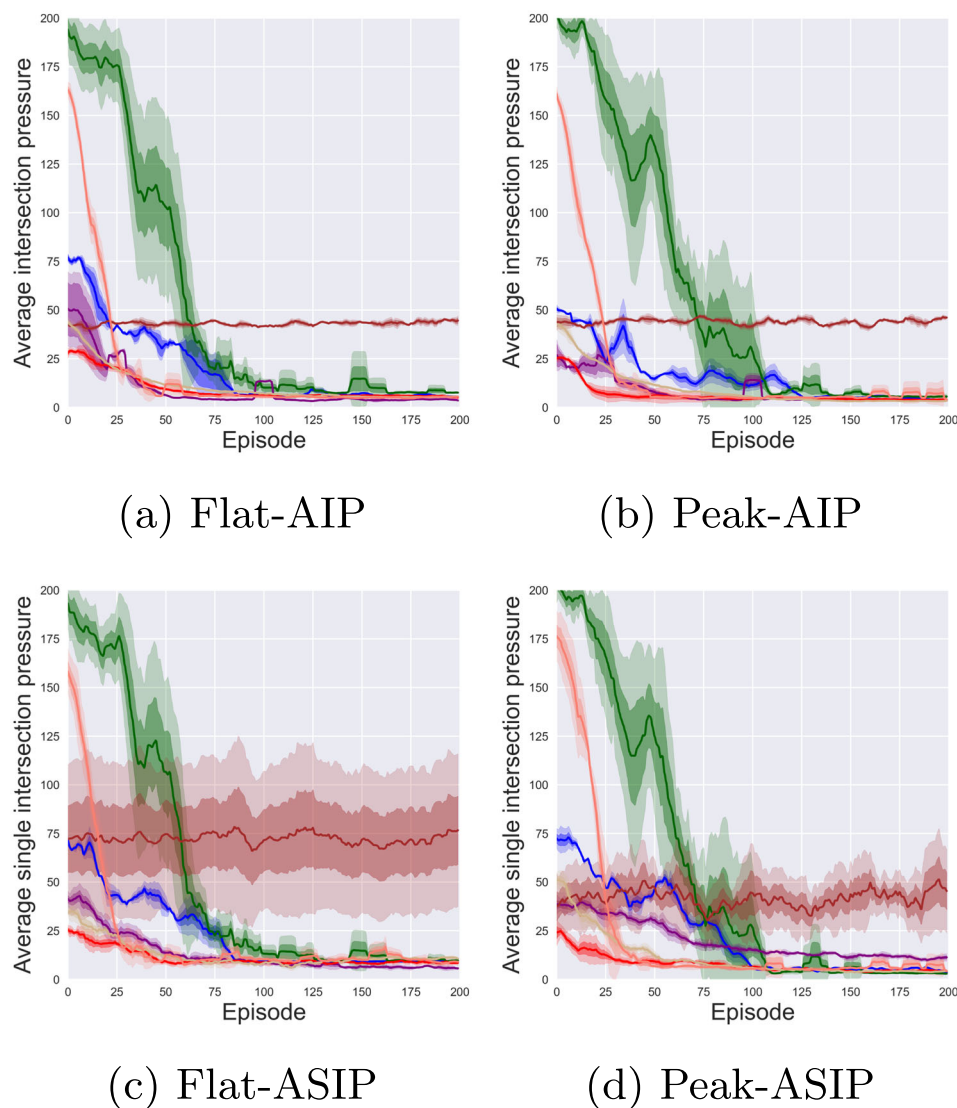


Fig. 10 Performance of different algorithms in AIP and ASIP metrics

Table 5 The results of arterial transportation network comparison are averaged by 5 times for each model

Models	Flat _{6×1}				Peak _{6×1}			
	ATT	AQL	AIP	ASIP	ATT	AQL	AIP	ASIP
FixedTime	149.364	14.304	×	×	162.515	16.640	×	×
MaxPressure	99.408	4.127	×	×	99.660	4.015	×	×
SOTL	117.217	7.242	×	×	120.396	7.673	×	×
GA	252.973	32.978	40.474	66.069	327.983	43.721	40.928	30.184
PressLight	98.467	3.780	5.613	7.984	100.332	3.903	4.293	4.690
CoLight	96.161	3.286	7.324	9.314	111.679	6.337	5.309	3.734
MpLight	96.803	3.332	5.198	7.991	98.139	3.755	3.642	3.724
AttentionLight	100.435	3.550	5.587	9.007	316.896	6.436	3.672	3.657
DiffLight	93.261	3.012	4.914	6.372	95.759	4.120	3.337	3.092
DiffusionLight	92.329	2.968	4.587	6.691	93.027	3.104	3.035	2.865

based on the DM performs well and DiffusionLight uses the shortcut theory to obtain information and achieve better performance through Encoder.

5.6 Data missing performance comparison

We train and test our method on online datasets with different missing rates and compare our method to all baseline methods. DiffusionLight performs best on most datasets. Observations of the environment simulate real-world data missing through random disabling.

We show the effect of different models with different data missing ratios in Fig. 11. In the Peak_{6×1} experiment, the stability of CoLight and PressLight is greatly challenged when the missing ratio increases, compared to the other models our model DiffusionLight remains stable under faster convergence.

In the observation data random missing experiments, we can see that DiffusionLight achieves optimal or sub-optimal performance compared to other baseline methods in Table 6. The method based on DMs helps to alleviate the influence of external factors by generating actions through multiple noise predictions. The performance of other baseline methods is more disturbed by observations and returns than methods based on diffusion processes.

We evaluated the generalization performance of DiffusionLight on Flat_{6×1} and Peak_{6×1} at different missing rates. We first trained at a specific missing rate and then tested at other missing rates. The generalization performance for different missing rates can be expressed as:

$$F_r = F_{train}/F_{test} \quad (32)$$

where F_r is the relative generalization performance for the current missing rate, F_{train} is the performance of models trained at other missing rates, and F_{test} is the performance of models trained at the current missing rate. As Fig. 12 shows, DiffusionLight performs well in the most situations. When

we train DiffusionLight at low missing rates and test it at high missing rates, the performance drops. This is because in the case of high missing rates, the model faces more severe missing situation, making it difficult for the model to work out.

5.7 Ablation experiment

In this paper, we propose a stable and convenient diffusion model that requires the model to maintain excellent performance in both normal and missing data, and this subsection summarizes the above as the following three issues:

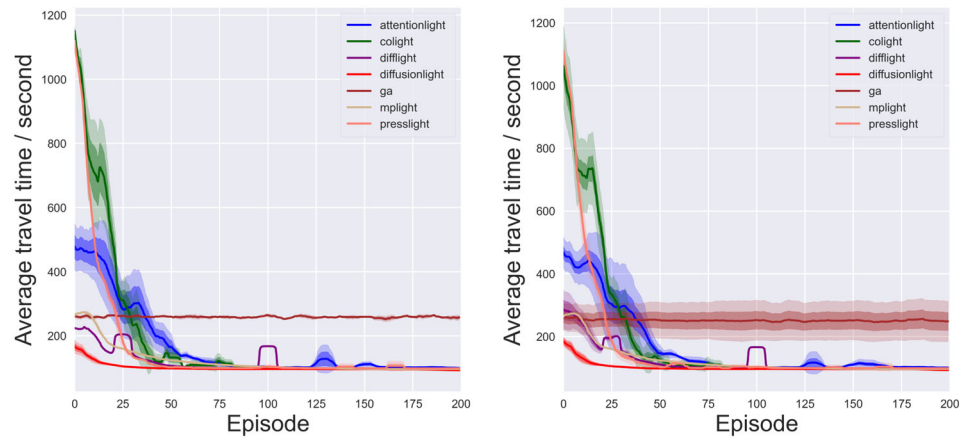
- 1) Is it effective to use shortcut theory in RL algorithm with DM?
- 2) The effect of diffusion steps N' on the shortcut-diffusion model?
- 3) Impact of different modules under missing data experiments?

In order to answer the three questions above, we use four of the methods shown in Table 7. Where (-SEM) does not include the anomalous state estimation models.

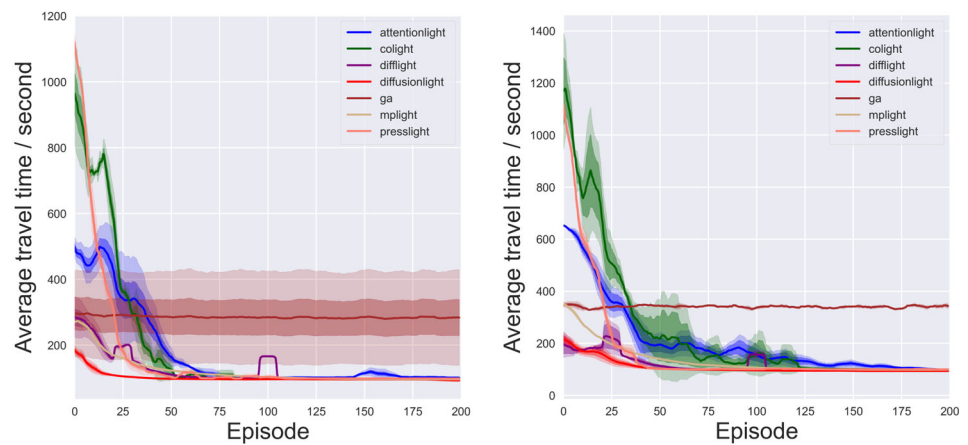
We first show the effect of shortcut theory and the number of diffusion steps on the model in Fig. 13. In Fig. 13(a) and (b) DiffusionLight-10 both demonstrate certain effects compared to Masac and Masac-d50. Another concern is that the diffusion model with the addition of shortcuts does not necessarily achieve better convergence at higher iteration steps, suggesting that shortcut speeds up model fitting and that the results of the network that is too deep will be overfitted.

We show the role of different modules in data loss in Fig. 14, and we can find that the Masac-d model will obtain great convergence when the diffusion model is added. When we remove the neighbor state estimation module in data loss, the DL-SEM model will partially lose its ability to complement the anomalous data.

Fig. 11 Performance of different algorithms in different missing rates



(a) 5% rate in Flat_{6×1} (b) 15% rate in Flat_{6×1}



(c) 30% rate in Flat_{6×1} (d) 5% rate in Peak_{6×1}

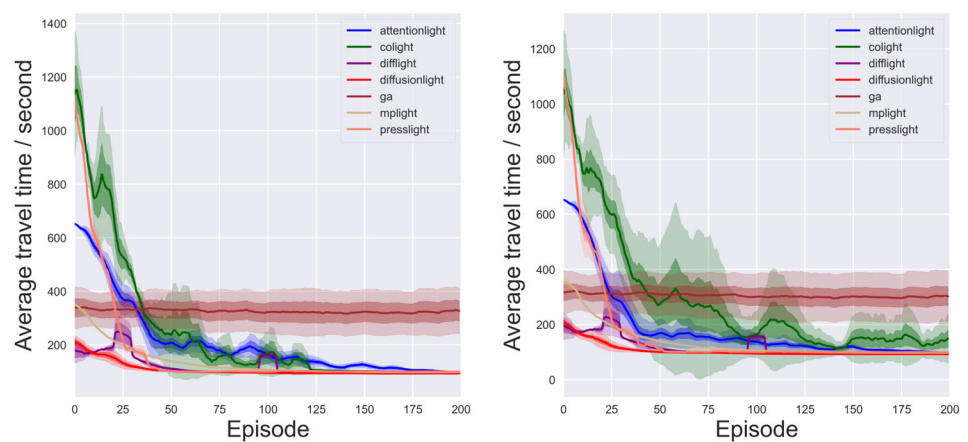
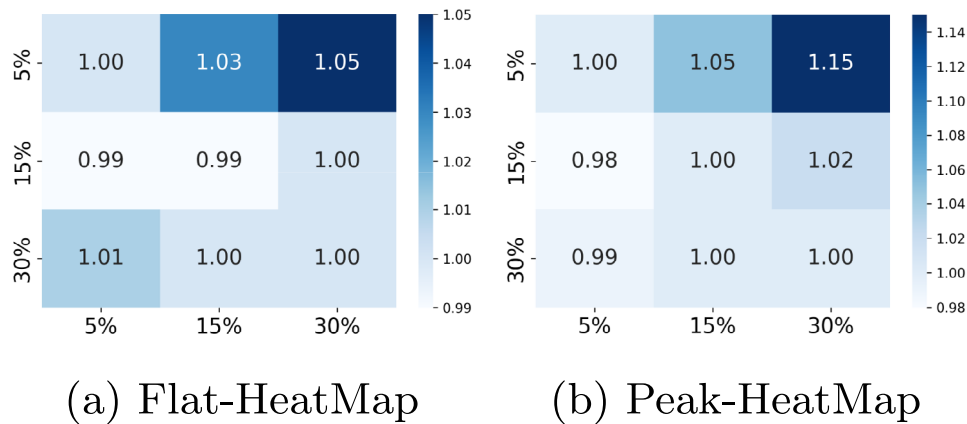
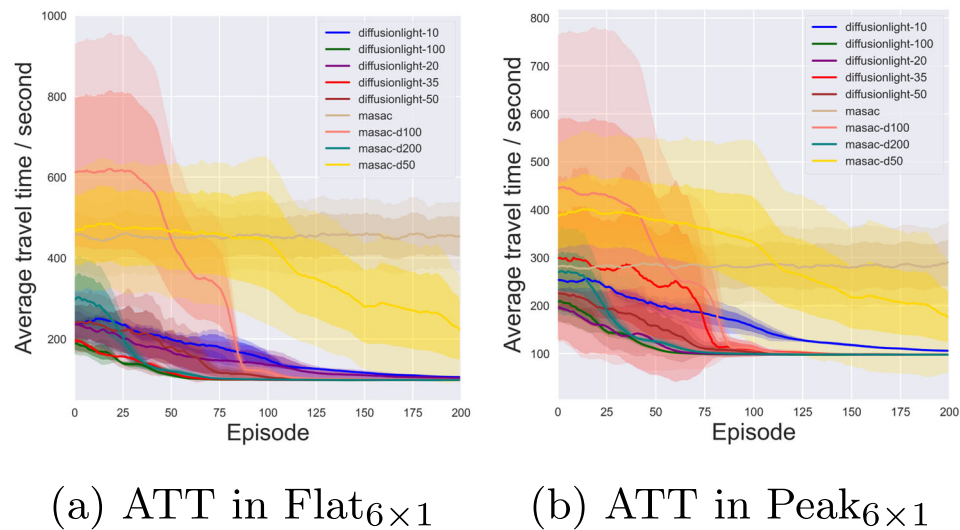


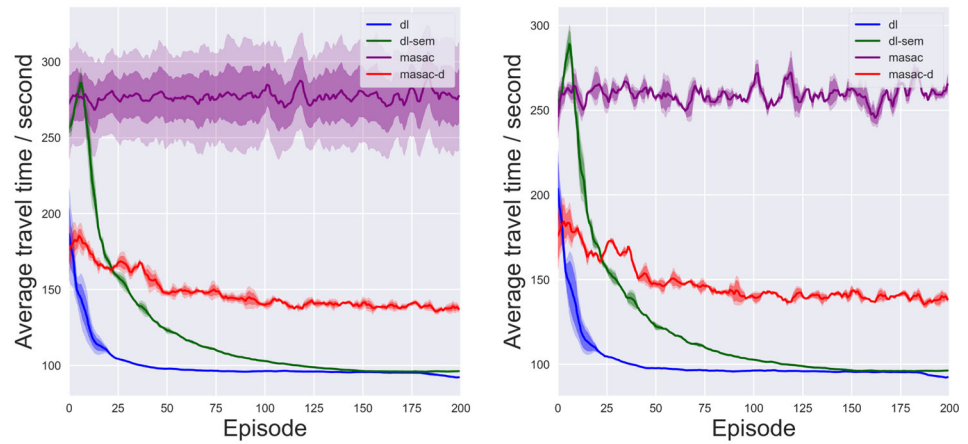
Table 6 The ATT metric of different algorithms at different data missing rates

Dataset	Rate(%)	FixedTime	GA	PressLight	CoLight	MpLight	AttentionLight	DiffLight	Ours
Flat _{6×1}	5	149.364	214.927	97.776	99.712	96.995	99.517	97.356	96.686
	15		238.844	104.435	99.827	97.997	99.372	97.927	97.643
	30		297.173	125.713	114.012	107.009	112.719	102.108	100.489
Peak _{6×1}	5	162.515	233.116	98.137	100.864	97.535	99.517	98.182	97.004
	15		245.134	112.003	109.902	102.921	110.006	100.689	99.339
	30		312.773	150.012	130.945	109.846	122.319	108.112	105.726

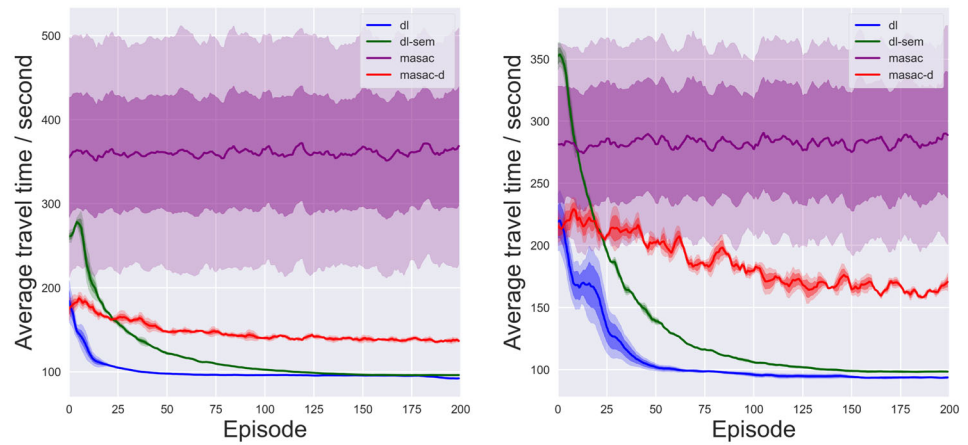
**Fig. 12** The relative generalization performance of DiffusionLight in different missing rates**Fig. 13** Performance of shortcut and different diffusion steps in ablation experiment**Table 7** The effect of shortcut theory and diffusion step number on the model

Models	CTDE	Diffusion	Shortcut	SEM
MASAC	✓			
MASAC_D	✓	✓		
DL(-SEM)	✓	✓	✓	
DL	✓	✓	✓	✓

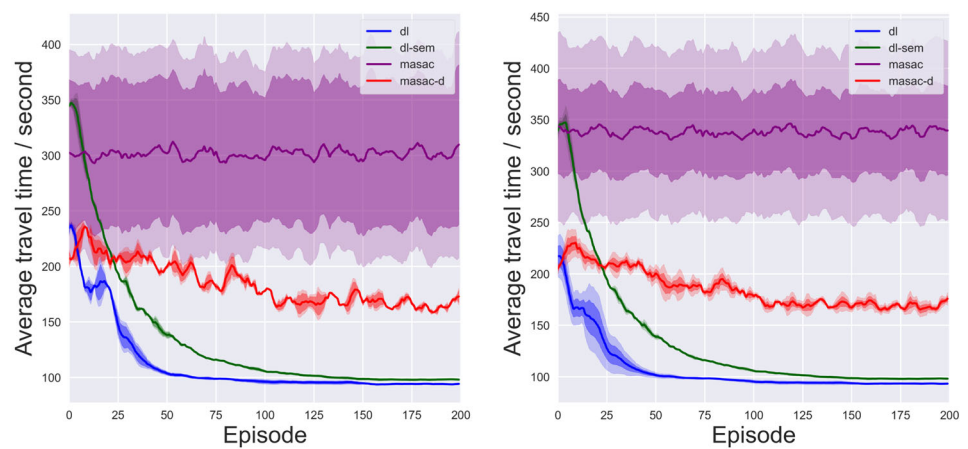
Fig. 14 Performance of diffusion modules in ablation experiment



(a) 5% rate in Flat_{6×1} (b) 15% rate in Flat_{6×1}



(c) 30% rate in Flat_{6×1} (d) 5% rate in Peak_{6×1}



(e) 15% rate in Peak_{6×1} (f) 30% rate in Peak_{6×1}

Table 8 The effect of shortcut theory and diffusion step number on the model

Models	Steps	task		TC(s)
		Flat _{6×1}	Peak _{6×1}	
Fixedtime	×	149.364	162.515	×
MASAC	×	301.192	276.123	12 ± 1.0
MASAC_D	50	180.163	183.472	112 ± 3.90
	100	93.445	93.595	227 ± 6.19
	200	92.741	93.036	454 ± 8.07
DL(-SEM)	10	96.218	101.783	14 ± 0.89
	20	93.379	96.009	16 ± 1.12
	35	92.967	93.104	22 ± 2.37
	50	92.779	93.190	135 ± 4.61
	100	92.706	93.070	242 ± 5.85

In the ablation experiments shown in Tables 8 and 9, we validate the impact of different modules under normal and anomalous data, respectively. As shown in Table 8 verifies that we add diffusion model in reinforcement learning still maintains excellent performance in huge state space, and with the addition of shortcut theory, the time cost is greatly ablated while maintaining stability. Meanwhile, Table 9 gives the effects of different modules under different data missing rates, which can be clearly reflected in the experiments of DL(-SEM) that the shortcut theory not only brings better collaboration ability but also maintains better performance under the same number of diffusion steps. When we adopt the state estimation model for anomaly data correction, the mechanical error of the actual traffic is mitigated to a certain extent.

Table 9 The effect of shortcut theory and diffusion step number on the model

Models	MR(%)	task		TC
		Flat _{6×1}	Peak _{6×1}	
Fixedtime	×	149.364	162.515	×
MASAC	5	268.992	274.127	12 ± 1.0
	15	245.151	293.643	
	30	351.901	330.423	
MASAC_D	5	135.531	147.525	20 ± 2.15
	15	142.363	158.645	
	30	224.118	165.314	
DL(-SEM)	5	96.992	97.512	21 ± 2.64
	15	97.771	99.772	
	30	101.372	106.992	
DL	5	96.686	97.004	22 ± 2.59
	15	97.643	99.339	
	30	100.489	105.726	

6 Conclusion

In this paper, we introduce Diffusionlight, a new method for collaborative TSC with multiple intersections. It employs the DM and centrally trained SAC entropy updates. We design the strategy network as a shortcut-diffusion model, which first uses global observations for shortcut jumps, and then the diffusion model utilizes local observations to perform noise reduction from new diffusion points and output actions. For the missing data problem, we design the data filling method which uses neighbor nodes for conservative data filling. In tests on multiple datasets, we demonstrate that DiffusionLight exhibits several advantages such as stability, fast convergence, and missing filling. Of course, we note that the cooperative relationship between agents is an implicit one, and the goodness of this cooperative approach relies on the quality of the data and the capability of the model. We are concerned that the Group-Aware Coordination approach manages agents in groups through explicit metrics, guiding agents with similar historical data to maintain consistent behavior through group cohesion. In future work, we will think about how to apply the idea of Group-Aware Coordination in TSC problems to further improve the model's ability to cope with multiple scenarios by enhancing the consistency of the agents within a group and the cooperation between groups.

Appendix A: Section title of first appendix

caption 1 For all x and y in $\mathcal{R}^n$, if there is a real number $0 \leq \lambda < 1$, the function $f : \mathcal{R}^n \rightarrow \mathcal{R}^n$ becomes a compression function:

$$\|f(x) - f(y)\| \leq \lambda \|x - y\| \quad (\text{A1})$$

According to the mean value theorem $f(x) - f(y) = f'(k)(x - y)$, $y < k < x$, we can know that when $f(x)$ satisfies the (A2), then $f(x)$ shrinks at $0 \leq \lambda < 1$ times the speed.

$$\sigma_{\max} \left(\frac{\partial f(x)}{\partial x} \right) \leq \lambda < 1 \quad (\text{A2})$$

Where $\sigma_{\max}(A)$ represents the largest singular value of matrix A . The compression mapping in Note is closely related to the Lipschitz continuity, and in fact the function satisfying (A1) and $\lambda > 0$ is called the λ -Lipschitz continuous function.

Proof Equation (A2). For matrix A , its maximum singular value is defined as $\sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)}$, where $\lambda_{\max}(A^T A)$ is the largest eigenvalue of $A^T A$.

Consider a small increment h , take $y = x + h$, and bring into (A1) to obtain:

$$\|f(x + h) - f(x)\| \leq \lambda \|h\| \quad (\text{A3})$$

Consider the definition of matrix derivatives and expand the above equation to obtain:

$$\left\| \frac{\partial f(x)}{\partial x} h \right\| \leq \lambda \|h\| \quad (\text{A4})$$

Equation (A4) shows that matrix $\frac{\partial f(x)}{\partial x}$ satisfies the condition for any vector h . Thus, (A2) can be obtained.

Now this paper gives the discrete stochastic compression theorem, which is an improvement of the stochastic difference equation compression theorem in references [22]. $\square$

caption 2 [22] considers a random difference equation in the form of (A5) :

$$a^{i+1} = f(a^i, i) + g(a^i, i)\epsilon^i \quad (\text{A5})$$

$f(\cdot, i)$ is a mapping function of $\mathcal{R}^n \rightarrow \mathcal{R}^n$, for $i \in N, \dots, 0$, $g(\cdot, i)$ is an $\mathcal{R}^n \times N \rightarrow \mathcal{R}$ function and ϵ^i is an independent sequence of n -dimensional zero mean unit variance Gaussian noise vectors. Assume that the system satisfies the following two assumptions:

H1) For all $i \in N$, the function $f(\cdot, i)$ contracts at the rate of factor λ in the sense of (A2).

H2) The effect of noise is uniform by the constant: $\text{Tr}(g(a^i, i) \text{I} g(a^i, i)) \leq C, \forall a^i, i$

Then, for the two sample trajectories a^i and $\tilde{a}^i$ satisfying (A5), we have:

$$\mathbb{E} \|a^i - \tilde{a}^i\|^2 \leq \frac{2C\tau}{1-\lambda^2} + \lambda^{2i} \mathbb{E} \|a^0 - \tilde{a}^0\|^2 \quad (\text{A6})$$

A.1 Corollary 1

The reverse process of the conditional DM in this paper is expressed as:

$$\dot{a}^{i+1} = f(a^i, i) + g(a^i, i)\epsilon^i \quad (\text{A7})$$

$$a^{i-1} = X\dot{a}^{i+1} + b \quad (\text{A8})$$

Where $X \in \mathcal{R}^{n \times n}$ is a non-extended linear map, $f(a^i, i)$ and $g(a^i, i)$ both satisfy the assumptions (H1) and (H2). If (A8) is substituted into (A7), there is:

$$a^{i+1} = Xf(a^i, i) + b + g(a^i, i)X\epsilon^i \quad (\text{A9})$$

$$\tilde{a}^{i+1} = Xf(\tilde{a}^i, i) + b + g(\tilde{a}^i, i)X\epsilon^i \quad (\text{A10})$$

Define $\tilde{f}(a^i, i) = Xf(a^i, i) + b$, according to (A2):

$$\sigma_{\max} \left(\frac{\partial \tilde{f}(a^i, i)}{\partial a^i} \right) \leq \sigma_{\max}(X) \sigma_{\max} \left(\frac{\partial f(a^i, i)}{\partial a^i} \right) = \lambda \quad (\text{A11})$$

Where $\sigma_{\max}(X) \leq 1$ is a non-extended linear map. Then, you can get:

$$\text{Tr}(g(a^i, i)X^T X g(a^i, i)) = g(a^i, i) \text{Tr}(X^T X) = C\tau \quad (\text{A12})$$

Where $\tau = \frac{\text{Tr}(X^T X)}{n}$. Then,

$$\begin{aligned} & \mathbb{E} \|a^{i+1} - \tilde{a}^{i+1}\|^2 \\ &= \mathbb{E} \|\tilde{f}(a^i, i) + g(a^i, i)X\epsilon^i - \tilde{f}(\tilde{a}^i, i) - g(\tilde{a}^i, i)X\epsilon^i\|^2 \\ &= \mathbb{E} \|\tilde{f}(a^i, i) - \tilde{f}(\tilde{a}^i, i)\|^2 \\ &+ \mathbb{E} \|g(a^i, i)X\epsilon^i - g(\tilde{a}^i, i)X\epsilon^i\|^2 \\ &\leq \lambda^2 \mathbb{E} \|f(a^i, i) - f(\tilde{a}^i, i)\|^2 \\ &+ \mathbb{E} \|(g(a^i, i) - g(\tilde{a}^i, i))X\|^2 + \mathbb{E} \|\epsilon^i\|^2 \\ &\leq \lambda^2 \mathbb{E} \|a^i - \tilde{a}^i\|^2 + \mathbb{E} \|(g(a^i, i) - g(\tilde{a}^i, i))X\|^2 \end{aligned} \quad (\text{A13})$$

According to (H2), we can see:

$$\begin{aligned} & \|g(a^i, i) - g(\tilde{a}^i, i)\|^2 \\ &= 2C\tau - 2\|g(a^i, i)g(\tilde{a}^i, i)\| \\ &\leq \lambda^2 \mathbb{E} \|a^i - \tilde{a}^i\|^2 + 2C\tau \\ &\leq \lambda^{2(i+1)} \mathbb{E} \|a^0 - \tilde{a}^0\|^2 + 2C\tau \sum_{k=0}^i \lambda^{2k} \\ &\leq \lambda^{2(i+1)} \mathbb{E} \|a^0 - \tilde{a}^0\|^2 + 2C\tau \sum_{k=0}^i \lambda^{2k} \\ &\leq \lambda^2 \mathbb{E} \|a^i - \tilde{a}^i\|^2 + 2C\tau \\ &\leq \lambda^{2(i+1)} \mathbb{E} \|a^0 - \tilde{a}^0\|^2 + 2C\tau \sum_{k=0}^i \lambda^{2k} \\ &\leq \lambda^{2(i+1)} \mathbb{E} \|a^0 - \tilde{a}^0\|^2 + \frac{2C\tau}{1-\lambda^2} \end{aligned} \quad (\text{A14})$$

The above main results prove (A6).

A.2 Proof Theorem 1

In this paper, the training (A15) for noise prediction model ϵ_{θ_m} in the reverse process of diffusion model is given.

$$\mathcal{L}_{\epsilon}(\theta_m) = \mathbb{E}_{i \sim \mathcal{U}, \epsilon \sim \mathcal{N}(0,1), (s,a) \sim \mathcal{D}} [\|\epsilon_m - \epsilon_{\theta_m}(\sqrt{\alpha^i} a_m + \sqrt{1 - \alpha^i} \epsilon_m, s_m, i)\|^2] \quad (\text{A15})$$

Research shows that the noise prediction model ϵ_{θ_m} is a scaled version of the scoring function [28].

$$s_{\theta}(a, i) = -\frac{1}{\sqrt{1 - \alpha^i}} \epsilon_{\theta}(a, i) \quad (\text{A16})$$

Here we give the action output equation in the form of DDPM [9]. The reverse diffusion process is a random difference function, defined as (A17).

$$a^{i-1} = f(a^i, i) + \sigma^i \epsilon \quad (\text{A17})$$

Where $f(a^i, i) = \frac{1}{\sqrt{\alpha^i}} (a^i + (1 - \alpha^i) s_{\theta}(a^i, i))$ and $g^i = \sqrt{1 - \alpha^i}$, and $i = N', \dots, 1, 0$, then according to (A2), we can see:

$$\begin{aligned} & \frac{\partial f(a^i, i)}{\partial a^i} \\ &= \frac{1}{\sqrt{\alpha^i}} \left(I + (1 - \alpha^i) \frac{\partial s_{\theta}(a^i, i)}{\partial a^i} \right) \\ &= \frac{1}{\sqrt{\alpha^i}} \left(I - \frac{(1 - \alpha^i)}{1 - \bar{\alpha}^i} \right) I \\ &= \frac{1}{\sqrt{\alpha^i}} \frac{\alpha^i - \bar{\alpha}^i}{1 - \bar{\alpha}^i} I \\ &= \sqrt{\alpha^i} \frac{1 - \bar{\alpha}^{i-1}}{1 - \bar{\alpha}^i} I \\ &\leq \lambda < 1 \end{aligned} \quad (\text{A18})$$

The upper limit that can be determined by parameter a and therefore λ is:

$$\lambda = \max_{i \in \{N', \dots, 1\}} \sqrt{\alpha^i} \frac{1 - \bar{\alpha}^{i-1}}{1 - \bar{\alpha}^i} < 1 \quad (\text{A19})$$

From the stochastic difference function (H2), we can simply derive the constant C as:

$$C = n \max_{i \in \{N', \dots, 1\}} (1 - \alpha^i) \quad (\text{A20})$$

For the trajectory a^i and a^i of the action samples satisfying (A5), according to **caption 2**, the stochastic contraction inequality of reverse diffusion can be seen as follows:

$$\bar{\epsilon}^{N',0} \leq \frac{2C\tau}{1 - \lambda^2} + \lambda^{2N'} \bar{\epsilon}^{N'} \quad (\text{A21})$$

Where $\bar{\epsilon}^{N',0}$ represents the estimated error between reverse diffusion paths spreading downward to $i = 0$.

A.3 Proof Theorem 2

Continue scaling (A21), then:

$$\begin{aligned} \bar{\epsilon}^{N',0} &\leq \frac{2C\tau}{1 - \lambda^2} + \lambda^{2N'} \bar{\epsilon}^{N'} \\ \bar{\epsilon}^{j-1,0} &\leq \frac{2C^j\tau}{1 - (\lambda^j)^2} - (1 - \lambda^{2N'}) \bar{\epsilon}^{N'} \\ \bar{\epsilon}^{j-1,0} &\leq 2C^j\tau + (\lambda^j)^2 \bar{\epsilon}^{j,0} \end{aligned}$$

Where $j = N' - i$ and $\bar{\epsilon}^{j,0}$ represent the estimated error between reverse conditional diffusion paths that spread downward to j . So, we have:

$$\bar{\epsilon}^{N',0} \leq \bar{\epsilon}^{N'} \prod_{j=0}^{N'} (\lambda^j)^2 + \sum_{j=1}^{N'} \left(2C^j\tau \prod_{i=1}^{j-1} (\lambda^i)^2 \right) \quad (\text{A22})$$

Here λ^j and C^j are equally bounded by (A15) and (A16). Consider each item in (A22) separately, first according to **Definition 1**:

$$\begin{aligned} \bar{\epsilon}^{N'} &= (k^{N'})^2 \epsilon^0 + 2(b^{N'})^2 n \\ &= \bar{\alpha}^{N'} \epsilon^0 + 2(1 - \bar{\alpha}^{N'}) n \\ &= 2n + \bar{\alpha}^{N'} (\epsilon^0 - 2n) \\ &\leq 2n \end{aligned} \quad (\text{A23})$$

Continue to simplify the second part of (A22),

$$\begin{aligned}
 \prod_{j=1}^{N'} (\lambda^j)^2 &= \prod_{j=1}^{N'} \alpha^j \frac{(1 - \bar{\alpha}^{j-1})^2}{(1 - \bar{\alpha}^j)^2} \\
 &\leq \prod_{j=1}^{N'} \alpha^j \\
 &\leq \left(\frac{1}{N'} \prod_{j=1}^{N'} \alpha^j \right)^{N'} \\
 &= \left(1 - \frac{1}{N'} \prod_{j=1}^{N'} \beta^j \right)^{N'} \\
 &= \left(1 - \frac{\beta^{N'}}{2} \right)^{N'}
 \end{aligned} \tag{A24}$$

Where $\bar{\alpha}^j = \bar{\alpha}^{j-1} \alpha^j \leq \bar{\alpha}^{j-1}$, so the first step is simplified and the second step inequality is obtained by the arithmetic and geometric mean inequality. And finally, according to mathematical inequalities,

$$e^x \geq \left(1 + \frac{x}{N}\right)^N, \quad 0 \leq x \leq N, N \geq 1 \tag{A25}$$

Equation (A22) is finally simplified to:

$$\prod_{j=1}^{N'} (\lambda^j)^2 \leq e^{-\frac{N' \beta^{N'}}{2}} \tag{A26}$$

According to (A20), the constant C of DDPM is substituted into (A22) to simplify the third term:

$$\begin{aligned}
 &\sum_{j=1}^{N'} \left(2C^j \tau \prod_{i=1}^{j-1} (\lambda^i)^2 \right) \\
 &= \sum_{j=1}^{N'} \left(2n \max_{i \in \{N', \dots, 1\}} (1 - \alpha^i) \tau \prod_{i=1}^{j-1} (\lambda^i)^2 \right) \\
 &\leq 2n\tau \sum_{j=1}^{N'} \left(\beta^j \prod_{i=1}^{j-1} (\lambda^i)^2 \right) \\
 &\leq 2n\tau \sum_{j=1}^{N'} (\beta^j \cdot 1) \\
 &\leq 2n\tau N' \beta^{N'}
 \end{aligned} \tag{A27}$$

Finally, after (A23), (A26) and (A27) we can obtain:

$$\begin{aligned}
 \bar{\varepsilon}^{N',0} &\leq \bar{\varepsilon}^{N'} \prod_{j=0}^{N'} (\lambda^j)^2 + \sum_{j=1}^{N'} \left(2C^j \tau \prod_{i=1}^{j-1} (\lambda^i)^2 \right) \\
 &\leq 2n \cdot e^{-\frac{N' \beta^{N'}}{2}} + N' \beta^{N'} \\
 &\leq 2n \cdot \frac{\mu \varepsilon^0}{4n} + \frac{\mu \varepsilon^0}{2} \\
 &\leq \mu \varepsilon^0
 \end{aligned} \tag{A28}$$

The second inequality is simplified from the (A29) selection range.

$$\log \left(\frac{4n}{\mu \varepsilon^0} \right) \leq N' \beta^{N'} \leq \frac{\mu \varepsilon^0}{4n\tau}, \quad 0 < \mu < 1 \tag{A29}$$

The main result above proves that by **Theorem 2**, for any $0 < \mu < 1$, there exists $N' (N' < N)$ such that:

$$\bar{\varepsilon}^{N',0} \leq \mu \varepsilon^0 \tag{A30}$$

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Data Availability The data that support the findings of this study are openly available at <https://traffic-signal-control.github.io/>, reference number [reference number].

Declarations

Competing Interests All authors disclosed no relevant relationships.

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Code Availability <https://github.com/Y-ujl/DiffusionLight>

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